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Improving an Agent-Based Model of the UK Housing Market

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Abstract

Policymakers and forecasters worldwide require accurate, reliable and easily accessible tools to assist them in making informed decisions. One of these tools is an agent-based model of the UK housing market, which represents every household in the UK as an 'agent', each with assigned characteristics such as age, wealth, and income. This model is used to assess how different macroprudential policies, such as mortgage loan-to-income limits, may affect key groups in the housing market, such as first-time buyers or renters. I propose a variety of improvements to an existing agent-based model of the UK housing market, currently used by the Bank of England. I reduce the end-to-end run-time of large batches of simulation runs by 83.7% via implementing a model caching optimisation and parallelising batches of model runs. I create a stable validation metric to compare model accuracy, robustness and reliability on both a per-metric and per-model basis. Utilising this metric, I reduce the 2011 target-year model's composite validation loss by 7.79% through enhanced output calibration using the modern Trust Region Bayesian Optimisation method, whilst requiring 99.3% fewer model runs than the original model's calibration. I create a web-based graphical user interface for running experiments using this model, released as both an executable program and a publicly cloud-hosted website. Finally, I recalibrated the 59 of the model's 75 parameters for which publicly accessible data sources are available, to partially recalibrate the model to the state of the UK housing market in 2024. Through this variety of improvements across various parts of the model, I substantially improve the model's accessibility, diagnostic validation workflow, and computational performance to assist policymakers in making more informed, evidence-based, and robust policy assessments.

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Chapter 1

Introduction

In the UK, the 2008 Global Financial Crisis caused house prices to fall around 20% from the third quarter of 2007 to the second quarter of 2009, along with unemployment rates increasing from 5.2% in the first quarter of 2008 to 7.8% in the second quarter of 2009 [1]. Major global financial crises can be worsened by the housing market as evidenced in a working paper by the International Monetary Fund, which state that, 'the catalyst of the crisis was the overextended U.S. housing and mortgage markets' and then the 'trigger was the turnaround in U.S. house prices' [2, III. A.]. Additionally, the housing market is a particularly distinctive market because changes affect not just the individuals in a society, but also the major financial systems. In the UK, property wealth makes up the largest proportion (40%) of household wealth [3], so when the cost of mortgages and housing prices change, this affects individual households' wealth and debt, but also banks' capacity to lend and their asset quality. Because of this, housing is key for policymakers and regulators, who aim to target the highest-risk mortgages and the banks most exposed to them. These policymakers need to know which households have high debt relative to income and how many of these risky mortgages there are, so they can make policies that affect the riskiest cases without making mortgages less accessible to the majority. These policymakers clearly need tools that represent the diversity of households and the edge-targeting mortgage policies.

Agent-Based Models (ABMs) are simulations in which a system is modelled using many 'agents', each of whom make their own calculated decisions in an environment specific to the ABM type. These agents have their own parameters and rules and they interact with many other agents in an environment through a series of many small decisions. This leads to larger macro trends to emerge across a system. ABMs are often better suited to the analysis required by these housing-market policymakers than other traditional economic model types like Dynamic Stochastic General Equilibrium (DSGE) models and reduced-form models for several reasons [4]. Firstly, ABMs are well suited to scenarios, like the housing market, where there are many diverse characters in the playing field interacting. In the housing market, students have hugely different housing requirements from a retiree and are far more likely to be renting, due to lower income and total wealth. ABMs represent this heterogeneity far more effectively than these other model types because they do not force every agent into a small number of averaged groups. Macro trends are created directly through agent decisions and interactions, unlike DSGE and reduced-form approaches where the simulation is assumed to reach an equilibrium state, which does not suit the continual rise and fall in costs in the housing market. Secondly, ABMs are better for policy testing than these traditional models. Different policy changes affect separate groups in the housing sector differently. Rather than having one average household affected, you can see how policy changes affect these distinct groups. If policymakers are trying to make it easier for first-time buyers to buy houses, an ABM can show if a policy helps this group and if knock-on effects emerge elsewhere in the system, unlike traditional models, which report a single average response. Finally, ABMs naturally capture the boom-and-bust dynamics in the housing market. In the housing market, houses take time to build, it takes time for buyers/renters to find a house to move into and banks can suddenly reduce how much they are lending. Slight changes in policy can lead to a stronger market or a crash and ABMs can show the steps which lead to these reactions directly, unlike other models which assume the market will always settle towards an equilibrium.

Currently in the UK, mortgage lenders are restricted so that no more than 15% of their mortgages given out are at or above a 4.5 loan-to-income (LTI) ratio [5]. First-time buyers (FTBs) make up an average of 44% of mortgage volume, but they make up 54% of all high-LTI lending volume [6, fn. 15]. An example of a policy experiment that can be performed on an ABM of a housing market, is a tightening of this 15% limit to 10%. As these FTBs disproportionately rely on high LTI mortgages, fewer FTB agents would likely receive mortgage approvals. Policymakers could then directly quantify the effects of this policy shift on mortgage approvals, affordability and average debt-to-income, specifically for the first-time buyer group, which equilibrium models are less suited to doing.

Improving and recalibrating a full-scale ABM is challenging for three main reasons. Firstly, robust model validation and experimentation requires many repeated simulation runs across alternative parameter settings, stochastic seeds and scenarios, so even modest per-run costs become prohibitive at a batch scale [7, p. 367]. Secondly, output calibration is difficult because some behavioural parameters cannot be directly estimated from data and must instead be calibrated from noisy stochastic model outputs, turning calibration into an expensive black-box optimisation problem [8, Sec. 1]. Finally, recalibration requires recent, granular UK micro-data on households, wealth, credit, sales, rents and policy settings, but datasets are often delayed, confidential or commercially restricted, making a fully updated contemporary baseline difficult to construct.

This project aims to improve an existing ABM of the UK housing market, initially created by Baptista et al. in 2016 and later improved by Carro et al. in 2022 [9, 10], whilst collaborating with the Bank of England. I address these challenges in improving this ABM through a combined methodological, computational and empirical contribution. I introduce a common dimensionless validation loss which scores twenty heterogeneous housing- and mortgage-market targets on a comparable scale. I then use this loss as the objective for the Trust Region Bayesian Optimisation method [11], allowing the output-calibrated parameter space to be searched continuously rather than through a fixed grid. Alongside this, I develop a caching optimisation and parallel batch model run framework, and use these tools to support a 2024 recalibration of all model parameters for which suitable public data are available. Finally, I engineer a graphical user interface as both a Windows executable and as a cloud-hosted web application. Together, I improve the model's validity, runtime, transparency and practical accessibility. This project is best understood as an engineering and methodological improvement to an existing policy ABM, rather than as a completely new housing market ABM. The main contributions of this project are:

- Achieving a 6.13-fold increase in model run throughput on realistic experimental workloads, corresponding to a 83.7% runtime reduction.
- Reducing the model's validation loss by 7.79% whilst using 99.3% fewer model runs than the original model's calibration.
- Updating the 59 of the model's 75 calibrated parameters for which suitable accessible data sources are available to the 2024 target-year.
- Enabling users to visualise parameters, compare model versions, run experiments, and analyse results without command-line execution or bespoke post-processing scripts.

Chapter 2

Background

2.1 What is an ABM?

Agent-based models (ABMs) are a type of computational simulation, which aim to model the behaviour of a system. Uniquely to ABMs, overall system behaviour emerges from the actions made and interactions between many smaller entities in the system, each of which we call an *agent*. Examples of agents in different types of ABMs include people, locations, assets, institutions or households. Each of these agents holds their own state and makes their own decisions based on their own rules. This structure, where system-level outcomes are determined by smaller agents interacting within a system, is particularly useful when the individuals being modelled have diverse types, have local interactions between themselves or imperfect decision making [12, p. 7280].

A typical ABM consists of several core building blocks. Firstly, ABMs define what the agents actually are and what attributes or states these agents have. For example, if we are observing people as our agents, we might assign age, income or gender. Secondly, ABMs should have a defined environment in which the agents act and interact must be defined. For example, agents may act and interact with other agents in a market, through a network or in their local area. Thirdly, the behavioural rules for how agents decide, adapt to their environment and respond to information must be decided. For example, what do person or institutional agents do if they run out of money? Finally, an ABM must have a defined time structure. Modellers must define how agents are updated, if actions occur sequentially or simultaneously, how stochastic events are introduced, what the initial conditions are. The Overview, Design concepts and Details (ODD) protocol was developed to make these choices more transparent and reproducible by defining a standard structure to describe an ABM's purpose, entities, processes, assumptions and inputs [13, pp. 2763-2766].

For ABMs, calibration and validation are distinct but related stages of ABM development. Verification concerns whether the implemented code correctly represents the intended model and validation asks if the model is actually sufficient for its intended purpose [14]. Calibration is the process of estimating parameters so that simulated output aligns with empirical targets. This is difficult for ABMs in particular, as they often don't have an tractable or closed-form likelihood function. Often ABMs are calibrated by comparing simulated and observed real-world statistics, for example by using simulated minimum distance approaches [15]. Validation should then test whether the calibrated model reproduces empirical patterns, ideally using repeated runs, sensitivity analysis and comparisons across both micro-level mechanisms and macro-level outcomes.

A classic example of an ABM is Reynolds' flocking model to represent birds. In this model, the agents called 'boids' follow three simple local rules which are to avoid colliding with each other, match each others speed and stay as a group which causes the emergence of motion similar to a flock of birds in real-life [16]. More recently, Eubank et al. modelled contact networks for disease-transmission analysis using census and urban mobility data [17, p. 180]. They found that smallpox outbreaks could be contained through targeted vaccination combined with early detection at hotspots rather than requiring population-wide mass vaccination.

ABMs are useful in providing us with a way to study and understand complex systems from a simplified bottom-up approach. They allow us to represent differences between the individuals in a system, how these individuals act and observe how and why these system-level patterns emerge at an individual level. Moreover, ABMs are particularly useful to experiment on hypothetical scenarios where outcomes depend on how individual agents behave, interact and respond to each other over time.

2.2 Early stylised housing market ABMs

In 2009, Gilbert et al. created an early agent-based model, which happened to be a model of England [18]. Their aim was to create a model in which housing trends are created through interactions between buyers, estate agents, and sellers and not due to aggregates assigned in the model. This aligns with the concept that the housing market is very heterogeneous, and trends form through interactions of many different players, as explained in Chapter 1. In each period, individual households start out as either buyers or sellers. Local estate agents then provide valuations that sellers can use to set asking prices and buyers make offers on these prices. This creates a cycle of valuation, listing, search and buying which gives rise to trends in a variety of statistics, for example, average house price to earnings ratio.

Gilbert et al. met their aim to create a simple model of the English housing market, however, there are several key limitations of this early model. Firstly, their model scale, which has a maximum of only 2500 houses in their 'town', is far smaller than a model would need to be to predict the future of a large, full-scale housing market, like the whole of the UK. Secondly, the rental market was not a part of this model. The rental market is an important part of any housing market. In the UK, around 36% of households are rented privately or socially [19, Sec. 3], so incorporating these demographics of the population is a requirement to completely model the English housing market. Thirdly, calibration was done on a macro-level with parameters 'calibrated to match approximately the actual English situation' in 2008 [18]. This is a limitation as it is possible for the model to match macro-trends, like price-to-income ratio, on average, whilst having unrealistic searching and bidding driving it. Gilbert. et al even mention it 'may also be necessary to reproduce a real location in order to validate the model'.

Later, in 2014, Axtell et al. created the first large scale housing market ABM, which was of Metropolitan Washington D.C. [20]. They simulated Washington D.C. at full scale of over 2 million individual households as agents. This marks a clear increase in scale from Gilbert et al.'s small, stylised scale. They created this model in response to the Global Financial Crisis of 2008 and aimed to use their model to reproduce the crash in Washington D.C. and thereby explain why the crash occurred.

This Washington D.C. model is different and improved from Gilbert et al.'s simple model in several ways. Firstly, Axtell et al. combines many sources of transactional, characteristic, and administrative data to empirically calibrate many micro-components of the model. They then choose to validate the macro trends of the model against different datasets to ensure the model is producing 'correct' outputs. This is different to Gilbert et al.'s model in which there is limited empirical validation against real-world data. This is significant as Axtell et al. was the first major housing market ABM to explicitly use micro data for calibration. For my improvements to the UK housing market ABM, I will require validation from datasets within the UK. Although the model currently has some validation, I build upon this in Chapter 3 by developing a more systematic validation framework for comparing model versions against empirical UK housing-market targets. Secondly, Axtell et al. included renters in their model. Agents choose to buy a property if they can get a mortgage and if the house is within their budget. If they cannot buy a house they rent it. Moreover, agents can also become landlords by buying multiple houses and renting them out. This representation of the housing market is more realistic of a real-world property market.

Although Axtell et al. made several improvements from Gilbert et al.'s simple model, their model still has many limitations. The model does not correctly predict the bubble in neither size nor date and their rental market is underdeveloped. This is a clear limitation as the model does not quite fit its intended purpose. Perhaps this was a limitation of the input datasets, there are other external factors influencing the market during this era (the authors specifically mention seasonality and people moving house when schools start as examples of this) or the fact that the model doesn't use any location sensitivity.

Axtell et al. made major strides in developing a large scale ABM of a housing market and in their conclusion, they argue that their contributions to the field are just the beginning and they prove that this new model type is effective for building models of housing markets. This is still true today, as many modern housing market ABMs cite Axtell et al. for inspiration in their models.

2.3 Full-scale UK housing market ABMs

In 2016, Baptista et al. created the first full-scale model of the UK housing market, calibrated to reflect the state of the UK housing market in 2011, for the Bank of England. This was created in order to investigate the effect of a loan-to-income portfolio limit (capping how large a mortgage an individual can take out based on their income) and investigating the varying of the size of the buy-to-let market [9]. They found that making the buy-to-let market larger makes house prices more variable and makes market upswings and downswings more pronounced. This model is the first publication of the model I am improving for my individual project.

In 2022, Carro et al. published several improvements to Baptista et al.'s UK housing market model in order to conduct experiments for the Bank of England [10]. Their findings from the model are that, firstly, using both a hard loan-to-value limit and a soft loan-to-income limit reduce the house price cycle (booms and busts) by reducing credit availability. Secondly, policies targeting owner-occupiers often also carry effects to the rental sector as there is a shift from people being owner-occupants to becoming buy-to-let investors. Thirdly, there is lots of policy 'spillover' to different household types when enforcing policies, which gives evidence that the effect of policies needs to carefully be considered for all groups of people and not just the group of people the policy is aimed to target. This model is significant for my project as this is the version of the model I am improving for my project. The repository for this model can be found at `https://github.com/INET-Complexity/housing-model` and the last commit with code modification was in October of 2021, six months before Carro et al.'s working paper was published on the Bank of England's website.

This model uses households as the main agent type, each of which is an economic unit with their own wealth, income, age, and decisions. For example, each household decides whether they want to buy or rent when they currently do not live in a private house. It is easier to imagine a household as one or more individuals who live together and make collective decisions. Households make decisions through rule-based behavioural functions by responding to their characteristics, market conditions and policy constraints. Because these decisions are stochastic in nature, a fixed starting stochastic seed is used to make simulation runs reproducible and also allows us to evaluate how robust the model output is to stochastic variation.

As presented in Figure 2.1, there are four types of household agents in the model: renters, first-time buyers (FTBs), owner-occupiers and buy-to-let (BTL) investors. Although first-time buyers are effectively owner-occupiers (after they have bought a house), they are different in that they have no housing equity to roll-over when they first buy their house and hence, they are treated differently. When 'households' are born into the model they are considered to be in

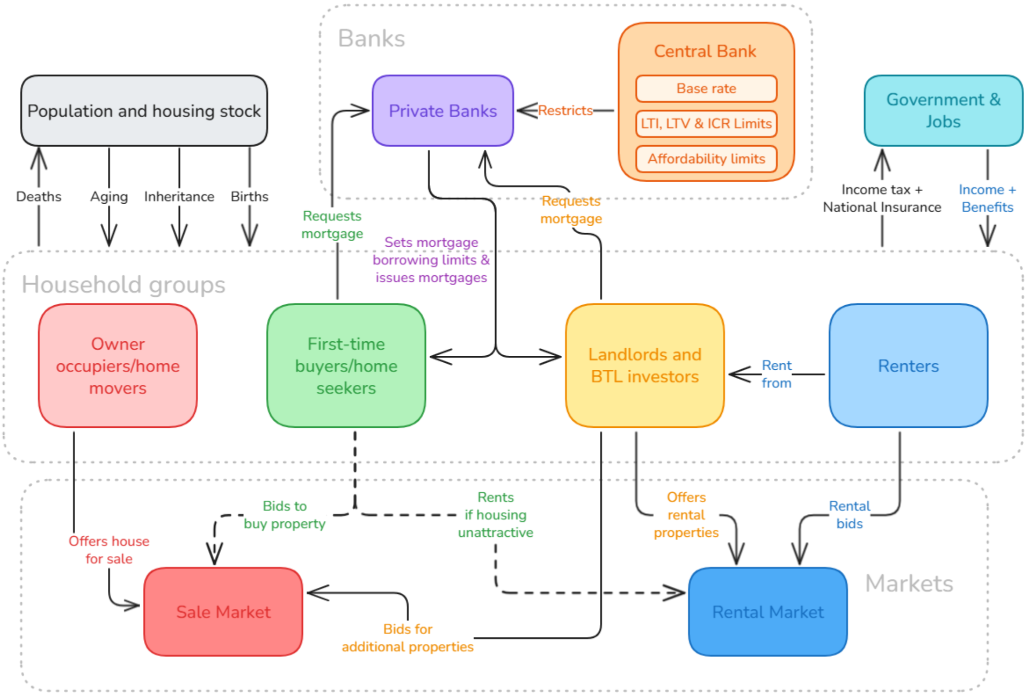

Figure 2.1: Structure of the UK agent-based housing market model

'social housing', which corresponds to a real-world state where the individuals in the household are homeless, living with parents or in temporary accommodation. Households can re-enter social housing when their rental contracts expire or they have just sold their house. Because new households must be 'born' into this initial social housing state, the model requires a 'spin-up' phase. This phase initially occurs with no households or houses in the model, followed up by a rapid build-up of population until the population reaches a target size, all in social housing. Then a house construction phase begins with rapid construction of the actual houses in the model. Households then make decisions about where to buy/rent until an equilibrium is reached, after which the model is considered initialised. This spin-up phase is mentioned to last approximately two hundred time steps in the model but I often run experiments discarding the first five hundred time steps.

The two other agent types in the model are banks and the central bank. The central bank defines the type of regulation used, if any, for mortgages. For example, the central bank may impose a limit on the amount a bank can loan compared with the value of a house (referred to as a hard loan-to-value ratio). The central bank is effectively the policymaker for this simulation. The bank, on the other hand, decides mortgage pricing and approves mortgages for households, based on the policies imposed by the central bank. The bank is a single aggregated lender and is not profit-incentivised.

Households go through a strict process when deciding to buy or let a house. When a household is in social housing for any reason, they must decide they want a (non-social) house. Households choose a desired expenditure budget, find the highest quality of house they can afford, decide whether to rent or buy a house and then place a bid on the sales market. If buying a house, households will pay fully in cash wherever possible, or if not, go through the bank to acquire a mortgage. This cycle repeats whenever a household is in social housing, which keeps a dynamic housing market going.

The first clear limitation of this model is scale. Although the model is said to model the entire population of ~27 million households, only 10,000 actual household agents are included in the model, so each household "agent" is representative of ~2,700 households in the UK. Baptista et al. state this is for computational reasons/complexity [9], but this could potentially cause a variety of issues. They explicitly state some housing quality bands have a limited number of transactions per month leading to a 'unrealistic distributions of price with quality'. I aim to improve the computational efficiency of this ABM to allow for simulations to efficiently be run at higher agent counts and I discuss my model performance enhancements further in Chapter 4.

Another limitation mentioned by Baptista et al. is that households defaulting on loans are not considered in the model. They explicitly state they ensure payments are made by giving bankrupt households cash in order to avoid foreclosure, which is not modelled. This is clearly unrealistic of a real-world housing market as it eliminates the possibility of foreclosure and lender losses.

A final limitation of this model is that no macro-economic dynamics are considered in this model; economic growth, recession, unemployment and population changes are not considered in this model. This means we are currently unable to use this model to consider how other economic factors affect the UK housing market. It would be interesting to see how a thriving economy influences the house prices, or if we are able to simulate/reproduce a housing market crash through unemployment shocks.

Carro et al. make a variety of improvements to this model, all of which are targeted at making the model more representative of the UK housing market, therefore making the model more reliable and accurate for running experiments and giving policymakers insights.

The first clear improvement to the model from Carro et al. is expanded use of micro-data. Carro et al. collect and utilise a much larger set of micro data from a variety of sources, such as Zoopla housing/rental listings and loan level regulatory data. Using a greater variety and quantity of micro data for calibration and validation allows use of full distributions, rather than just using averages, making the model better representative of the UK.

A second major improvement is their buy-to-let heterogeneity calibration. In Baptista et al.'s initial model, they assume a fixed distribution of whether a house owner becomes a buy-to-let investor across all income percentiles. This is not a great assumption, as logically those with higher incomes are more likely to look for investment opportunities. However, Carro et al. flips this by using survey data to estimate the chance of someone becoming a buy-to-let investor at the different income percentiles. Again, this improvement, alongside further model validation, makes the households in the model more representative of households in the UK, improving model reliability.

However, the limitations of this model are still similar to the limitations I mentioned for Baptista et al.'s initial model version. Carro et al. again explicitly mention the dynamics for defaulting on loans, not being able to vary macroeconomic factors and the down sampling of the model as clear limitations in the model.

Another limitation is this model's calibration period. Carro et al. state this model was calibrated to 2011 as this period is before the Bank of England's 2014 loan-to-income and affordability constraints [21, Ch. 5], 2011 is neither a clear boom nor a bust, and it also was said to have the strongest data availability in the period between the Great Recession and the introduction of the Bank of England's housing market policies [10, pp. 25-26]. This allowed Carro et al. to build a credible UK housing market baseline. However, in the 15 years subsequent to 2011, UK housing policy has shifted significantly since 2011, with landlord tax reforms altering buy-to-let incentives [22, Sec. 1] and this was followed by a fall in the buy-to-let share of new advances and a rise in first-time-buyer lending [22, Fig. 1] and it would be advantageous to explore if recalibrating the ABM to a more recent time period would reflect these changes. I discuss my contributions to recalibration in Section 5.2.

A further significant limitation I have currently noticed from working with this version of the model is accessibility and visualisation. The model output, similarly, to the original model, is a monthly time-series of core indicators, is presented as large comma-separated value (CSV) files. No aggregated or graphical visualisation for runs is currently included in the model, making insights harder to gain and see between runs. This means external scripts will need to be manually created to process and visualise data. For my project I personally want to make visualisation and comparison tools for this model, so it is easier for insights to be made and visualised quickly. We discuss our contributions to accessibility in Chapter 6.

2.4 Summary of housing market ABMs

In Table 2.1, I summarise the reviewed ABMs from Sections 2.2 and 2.3. I choose to combine Baptista et al. and Carro et al. and represent it as one model as Carro et al. improved Baptista et al.'s model and they both represent the same model at different time periods.

Model	**Scope**	**Calibration approach**	**Validation approach**	**Main limitation**
Gilbert et al. (2009) [18]	**Households:** 2,500 **Agents:** 2,500 **Location:** England **Target year:** 2008	Macro-style calibration to English 2008 conditions. Simple behavioural rules for buyers, sellers and estate agents.	Qualitative face validation through reproducing responses to shocks such as interest-rate, income and demand changes.	Small and stylised; no rental market, repossessions or forced sales, and no real-location validation.
Axtell et al. (2014) [20]	**Households:** ~2,000,000 **Agents:** ~200,000 **Location:** Washington, D.C. **Target year:** 1997	Behavioural components estimated from micro-data, including household survey data, multiple-listing-service transactions, housing characteristics and mortgage-servicing records.	Simulated outputs compared with historical housing-market time series, including price indices, sale prices, listings, transactions, days-on-market, months of inventory, homeownership and vacancy.	Does not fully match the housing bubble's timing or magnitude; rental-market, seasonality, location, construction and banking mechanisms remain underdeveloped.
Baptista et al. (2016) / Carro et al. (2022) [9, 10]	**Households:** ~27,000,000 **Agents:** 10,000 **Location:** United Kingdom **Target year:** 2011	Systematic full-parameter estimation and calibration: most parameters estimated from micro-data, some postulated with sensitivity analysis, and five behavioural parameters calibrated using simulated method of moments.	Compared simulated outputs with 2011 empirical values, 2005-2014 empirical ranges, stylised house-price cycles and distributional features across repeated Monte Carlo simulations.	Significant population downsampling (1:2700) creates sparse transaction issues; bankruptcy, default, foreclosure, monetary-policy and banking-sector dynamics are simplified.

Table 2.1: Comparison of the housing-market ABMs discussed in the background chapter.

Chapter 3

A target-metric loss function for validation

Validation provides the evidence that the model adequately represents the system it is attempting to model, which in our case is the UK housing market. Although validating any model is a significant and important task, validation is especially relevant for ABMs because they often contain complex mechanisms which generally represent more complex real-world systems [23, p. 1].

Currently, this model has been validated by empirical benchmarking; by comparing whether its simulated outputs resemble real UK housing and mortgage market data, both in aggregate form and across distributions. This includes comparing the simulated averages with 2011 data including house prices, transactions, mortgage approvals, debt-to-income, LTV and LTI ratios, borrower ages, risky lending composition and buy-to-let outcomes [10, Ch. 4]. The current validation for this model is robust to show that the model in its current form successfully produces the complex market outcomes across a variety of macro-level indicators and validation is also distribution-aware.

However, there are several limitations of this current method. Firstly, the validation remains partly qualitative. Although the model is shown to match several empirical features of the UK housing market, the strength of this fit is not always measured numerically. As a result, assessments such as whether the model matches the amplitude and frequency of house price cycles, or whether simulated distributions are sufficiently close to observed distributions, rely partly on visual inspection and researcher judgement. This makes it difficult to assess precisely how well the model validates overall, or to compare whether one version of the model represents an improvement over another.

Secondly, the validation metrics are assessed largely independently. The model is compared against a broad set of indicators and although this breadth is necessary, the current methodology does not provide a systematic way of aggregating these comparisons or evaluating trade-offs between them. For example, a change to the model may improve its fit to mortgage approvals while worsening its fit to housing transactions or borrower-level distributions. Without a common framework for evaluating these discrepancies, it is difficult to determine whether the model has improved overall.

Thirdly, the current validation demonstrates that the model can reproduce several important empirical regularities of the UK housing market, but it provides less guidance on where the model fits poorly, which mechanisms are responsible for those discrepancies, or which areas should be prioritised during recalibration. This limits the usefulness of validation as an iterative model-development tool. A more systematic validation methodology would therefore strengthen the model by making empirical fit more transparent, comparable and informative across the full set of housing and mortgage market indicators.

3.1 Improving ABM validation via a target-metric loss function

To address these limitations, in this section I introduce a systematic, quantitative validation methodology. I first calculate twenty real-world macro-level indicators, which I define as our 'validation target metrics'. All of the data sources for these metrics are defined in appendix table B.1. Let $\mathcal{S}$ denote the set of required scored validation metrics. These twenty metrics are calculated for both the target years of 2011 and 2024 respectively. The 2011 validation target metrics are used to validate the original housing market calibration [10] with the target year of 2011. The 2024 validation target metrics are used to analyse and validate the recalibration of the model to the 2024 target year, which I discuss later in Chapter 5. By calculating equivalent metrics for both the 2011 and 2024 target years, I allow direct comparison of model fit to validation targets across calibration years. I define all twenty of these validation metrics in Table 3.1 alongside the family they belong to, which are defined in Subsection 3.1.2.

Metric	**Family**
Mortgage Approvals	**pos**
Housing Transactions	**pos**
Advances to First Time Buyers	**pos**
Advances to Home Movers	**pos**
Advances to Buy-to-let Investors	**pos**
Household Debt-to-income Ratio	**pos**
House Price-to-income Ratio	**pos**
House Price Growth	**add**
House Price Index Mean	**pos**
House Price Index Standard Deviation (Amplitude)	**pos**
House Price Index Cycle Period (Frequency)	**pos**
Rental Price Index Mean	**pos**
Household Owning Share	**share**
Household Private Renting Share	**share**
Owner Occupier Debt-to-income Ratio	**pos**
Rental Yield	**pos**
Interest Rate Spread	**add**
Income Distributional Realism	**low**
Housing Wealth Distributional Realism	**low**
Financial Wealth Distributional Realism	**low**

Table 3.1: All validation metrics alongside their respective families

To validate a particular model output, I chose to run 10 simulations, each for 3500 time steps, whilst discarding the first 500 time steps in order to allow the model to spin-up and stabilise. This is the same model setup used by Carro et al. in their validation of this model [10, Ch. 4]. For each validation metric $x \in \mathcal{S}$ (for example, mortgage approvals), let $\hat{x}_i$ denote the simulated value of that metric obtained from running the simulation from seed i. For example, if x is mortgage approvals, $\hat{x}_1$ would be the mean monthly mortgage approvals when running the model with a starting seed of 1. Let $\mathcal{I}_x$ be the set of validation seeds for which metric x is observed, with $n_x = |\mathcal{I}_x|$. For our validation runs, we use the starting seeds from one to ten in every case for simplicity and consistency. We therefore have $\mathcal{I}_x = \{1, \ldots, 10\}$ and $n_x = 10$ for every x in $\mathcal{S}$.

We define a validation target metric's 'target value' for all validation target metrics as the calculated macro-level indicator of values in that target year or the original distribution for distribution based metrics. We define a validation target metric, x's target value, as s_x. For each of these metrics we then define a 'target range'. We define this target range as the ideal range the simulation output will lie within, which is inspired by Neuländtner et al.'s 'acceptable target ranges', where an acceptable lower and upper bound is defined for each calibration criterion [24].

These target ranges are defined in B.2 and are defined by the source year range in the data, where possible, or otherwise as $s_x \pm 5\%$. Let us denote the validation target range for validation target metric x by $[L_x, U_x]$. Both the validation target metric's source value and ranges are defined in appendix table B.2.

3.1.1 Validation loss metric aims

Now we have defined our target metrics, values and ranges, it is important to define the different characteristics I want this metric loss to carry:

1. The loss function should place these twenty heterogeneous validation target metrics on a common, dimensionless scale, so that metric-level losses can be compared directly.

2. The primary contribution to each metric loss should reflect the accuracy of the model's central tendency.

3. The loss function should penalise stochastic unreliability by increasing loss when a large proportion of independent seed runs fall outside a target band, even if the seed mean is close to the target.

4. The loss function should penalise excessive between-seed variability, since a calibrated ABM should reproduce empirical targets robustly rather than only under favourable random seeds.

3.1.2 Validation metric families

Because the validation set contains several heterogeneous model outputs, a single raw distance measure is not appropriate for every metric. Counts, ratios, percentages, signed rates, and distributional divergence statistics have different domains and different interpretations of error. I therefore assign each validation metric $x \in \mathcal{S}$ to a loss family, each of which are calculated differently. The family each metric belongs to is again defined in Table 3.1. Let

$$\mathcal{F} = \{\mathbf{pos}, \mathbf{low}, \mathbf{share}, \mathbf{add}\} \tag{3.1}$$

1. **pos**: The **positive-level** family of metrics that are strictly positive and for which proportional errors are more meaningful than absolute errors. This includes counts, flows, price indices, ratios, yields, and other level variables. Thirteen out of the twenty target metrics we use in the model are positive level metrics. For these metrics, a model output that is twice the target should be treated comparably to one that is half the target. We will therefore use a log-ratio distance to the nearest edge of the validation band.

2. **add**: The **signed-additive family** is used for metrics that may naturally be close to zero and cross zero, which includes house price growth rates and interest-rate spread in our model. These metrics are instead scored using an additive distance, normalised by a robust metric-specific scale.

3. **share**: The **bounded-share family** is used for percentage-share metrics, such as household tenure shares. These metrics have a natural bounded domain of 0% to 100%, so errors are normalised by the full percentage-point domain rather than by the width of the target band. This prevents narrow target ranges from dominating the composite loss.

4. **low**: The **bounded-low-is-better** family is used for distributional divergence metrics. This includes our income, housing wealth and financial wealth distributional realism. These metrics are bounded by zero, indicating exact match between the simulated and empirical distributions. Since lower values are intrinsically preferable, the loss rewards improvements

within the acceptable band rather than only penalising values outside it. For a seed i, $\hat{x}_i$ is the Jensen-Shannon divergence (JSD) between the simulated household distribution and the corresponding empirically calculated Wealth and Assets Survey target histogram. I chose to use JSD over Kullback divergences, as it does not require absolute continuity and is finite and bounded, making distributional discrepancies more robust and interpretable for the sparse household categories [25, p. 145].

3.1.3 Explaining the generic metric loss formula

Let us define a generic formula for a metric's loss:

$$\mathcal{L}_x^{\mathcal{F}} = \underbrace{D_x}_{\text{central accuracy}} + \underbrace{W_R \cdot (1 - r_x)}_{\text{target-band reliability}} + \underbrace{W_V \cdot V_x}_{\text{between-seed dispersion}} \tag{3.2}$$

with weights $W_R = 0.5$ and $W_V = 0.25$, which are discussed below.

D_x The term D_x is a dimensionless measure of the distance between the simulated seed mean value $\bar{\hat{x}}$ and the target range $[L_x, U_x]$. The exact transformation used to compute D_x depends on which of the metric families, of which are described in section 3.1.4. We define which metrics belong to which metric family in table B.2. D_x receives unit weight and is the dominant component of the loss. For all families it measures the central accuracy of the model by comparing the seed-averaged simulated output with the empirical validation band. This ensures that the loss primarily rewards models whose expected behaviour is empirically plausible. For the bounded-low-is-better family, the distance term is deliberately not set to zero inside the acceptable band as lower divergence remains preferable even when the simulated distribution is within the acceptable threshold.

$(1 - r_x)$ The term $(1 - r_x)$ measures the proportion of validation seeds whose simulated output lies outside the target band and this term penalises unreliability with respect to the validation band. Even if the seed mean is close to the target range, a model is less robust if many individual seeds fall outside the acceptable range. This term therefore captures the probability that the model generates empirically implausible outcomes under stochastic repetition.

V_x The term V_x is a dimensionless transformation of the inter-seed variation. This penalises excessive dispersion across independent simulation seeds. This is important because an ideal ABM calibration should not reproduce empirical targets only under favourable random realisations; it should do so robustly across repeated runs. For most families we use the interquartile range to provide a robust measure of between-seed variability that is less sensitive to extreme seed outcomes than using the full range.

W_R, W_V The terms W_R and W_V are the relative weights given to the target-band reliability and between-seed variability components of metric loss respectively. These weights are intended to encode a priority ordering of central accuracy first, target-band reliability second, and between-seed dispersion third and hence we must choose $1 > W_R > W_V$. I chose W_R as 0.5 and W_V as 0.25 in this model as the natural choices for these parameters, with the unit weight on D_x to ensure that central accuracy remains the dominant contribution to the metric losses. I demonstrate the comparative robustness of these parameters in Section 3.2.

Together, these components combine central accuracy, robustness across seeds, and empirical reliability, while using family-specific normalisations to make heterogeneous validation metrics comparable within a single composite loss, which align closely with my per-metric loss aims defined in section 3.1.1.

3.1.4 Computing per-metric-family losses

Using the generic loss formula we define above in section 3.1.3, we can then define the loss metrics for each of our 4 defined metric families in section 3.1.2. Let us first define the simulated value seed-mean and the inter-quartile range (IQR) of the simulated metric values across seeds as:

$$\bar{\hat{x}} = \frac{1}{n_x} \sum_{i \in \mathcal{I}_x} \hat{x}_i, \qquad \mathrm{IQR}_{\hat{x}_i} = Q_{0.75}(\{\hat{x}_i : i \in \mathcal{I}_x\}) - Q_{0.25}(\{\hat{x}_i : i \in \mathcal{I}_x\}), \tag{3.3}$$

Let us also define the proportion of seeds whose simulated metric value lies inside the validation band as:

$$r_x = \frac{1}{n_x} \sum_{i \in \mathcal{I}_x} \mathbf{1}\{L_x \le \hat{x}_i \le U_x\}. \tag{3.4}$$

Let us also define the distance from the simulated value seed-mean ($\bar{\hat{x}}$) to the nearest edge of the validation band as:

$$d_x = \begin{cases} 0, & \text{if } L_x \le \bar{\hat{x}} \le U_x, \\ \min\big(|\bar{\hat{x}} - L_x|,\ |\bar{\hat{x}} - U_x|\big), & \text{otherwise.} \end{cases} \tag{3.5}$$

Let us also define a robust characteristic scale used for normalisation as:

$$A_x = \max\left\{ \underbrace{|s_x|}_{\text{source value magnitude}}, \underbrace{|L_x|,\quad |U_x|}_{\text{target range magnitudes}}, \underbrace{\frac{U_x - L_x}{2}}_{\text{target range half-width}}, \underbrace{10^{-9}}_{\text{zero-floor scale}} \right\}, \tag{3.6}$$

A_x makes signed-additive distances robust by normalising metrics to the largest relevant magnitude available, with the small floor preventing unstable blow-ups for targets with close-to-zero target ranges and source values.

Metric loss for the positive-level metric family

For the positive-level metric family (**pos**), we first define a distance component, in log-space, as:

$$D_x^{\mathbf{pos}} = \begin{cases} 0, & \text{if } L_x \le \bar{\hat{x}} \le U_x, \\ \ln\left(\frac{\bar{\hat{x}}}{U_x}\right), & \text{if } \bar{\hat{x}} > U_x, \\ \ln\left(\frac{L_x}{\bar{\hat{x}}}\right), & \text{if } 0 < \bar{\hat{x}} < L_x, \\ \ln(100), & \text{if } \bar{\hat{x}} \le 0. \end{cases} \tag{3.7}$$

We also define this family's variability component in log-ratio space as:

$$V_x^{\mathbf{pos}} = \begin{cases} \max\left(0, \ln\left(\frac{Q_{0.75}(\hat{x}_i)}{Q_{0.25}(\hat{x}_i)}\right)\right), & \text{if } Q_{0.25}(\hat{x}_i) > 0 \text{ and } Q_{0.75}(\hat{x}_i) > 0, \\ \frac{\mathrm{IQR}_{\hat{x}_i}}{A_x}, & \text{otherwise.} \end{cases} \tag{3.8}$$

Defining this family's distance and variation in log space allows errors to be treated proportionally rather than additively. For example, a rental yield which is three times too high will be assigned the same distance as one which is three times too low. This is particularly meaningful for these strictly positive housing-market quantities such as counts, ratios and indexes where relative error is more meaningful than raw absolute error [26, Secs. 1-3.3]. We can then finally define the positive-level metric loss to be:

$$\mathcal{L}_x^{\mathbf{pos}} = D_x^{\mathbf{pos}} + W_R \cdot (1 - r_x) + W_V \cdot V_x^{\mathbf{pos}}. \tag{3.9}$$

with weights $W_R = 0.5$ and $W_V = 0.25$.

Metric loss for the signed additive family

For the signed-additive family (**add**) of metrics defined in section 3.1.2, we calculate the loss as:

$$\mathcal{L}_x^{\text{add}} = \frac{d_x}{A_x} + W_R \cdot (1 - r_x) + W_V \cdot \frac{\text{IQR}_{\hat{x}_i}}{A_x}. \tag{3.10}$$

with weights $W_R = 0.5$ and $W_V = 0.25$.

Metric loss for the bounded-share metric family

For the bounded-share metric family (**share**), which have the natural bounds of $0 \le L_x < U_x \le 100$, loss is calculated accordingly via:

$$\mathcal{L}_x^{\mathbf{share}} = \frac{d_x}{100} + W_R \cdot (1 - r_x) + W_V \cdot \frac{\text{IQR}_{\hat{x}_i}}{100} \tag{3.11}$$

with weights $W_R = 0.5$ and $W_V = 0.25$.

Metric loss for the bounded-low-is-better metric family

For the bounded-low-is-better metric family (**low**), we first define a distance component as:

$$D_x^{\mathbf{low}} = \begin{cases} 0.25\,\dfrac{\bar{\hat{x}}}{U_x}, & 0 \le \bar{\hat{x}} \le U_x, \\ 0.25\,\dfrac{\bar{\hat{x}}}{U_x} + \dfrac{\bar{\hat{x}} - U_x}{U_x}, & \bar{\hat{x}} > U_x, \end{cases} \qquad U_x > 0,\ \bar{\hat{x}} \ge 0. \tag{3.12}$$

This applies a smaller proportional penalty when divergence is within the acceptance range, but adds a full penalty once $\bar{\hat{x}} > U_x$. This allows a metric with a JSD of 0.10 to be reflected as less accurate than a metric with a JSD of 0.05. We can then define the overall loss for this family as:

$$\mathcal{L}_x^{\mathbf{low}} = D_x^{\mathbf{low}} + W_R \cdot (1 - r_x) + W_V \cdot \frac{\text{IQR}_{\hat{x}_i}}{U_x}. \tag{3.13}$$

with weights $W_R = 0.5$ and $W_V = 0.25$.

3.1.5 Computing overall loss

Let $\mathcal{S}$ denote the set of required scored validation metrics, and let w_x denote the metric weight. The overall composite validation loss is the weighted mean:

$$\mathcal{L}^{\text{composite}} = \frac{\sum_{x \in \mathcal{S}} w_x \mathcal{L}_x}{\sum_{x \in \mathcal{S}} w_x}. \tag{3.14}$$

Although in the current validation implementation, for simplicity, all scored metrics have $w_x = 1$, so Equation 3.14 reduces to the arithmetic mean of the required metric losses, this framework still gives capability to assign higher weights to metrics deemed more important for calibration.

3.2 Sensitivity analysis of validation loss weights

As defined in Equation 3.2, the weights W_R and W_V in our loss are intended to encode a priority ordering rather than to represent empirically estimated coefficients. The unit weight on D_x ensures that central accuracy remains the dominant contribution to each metric loss, since the primary aim is to reward models whose seed-averaged behaviour is close to the empirical validation band. The weights of $W_R = 0.5$ and $W_V = 0.25$ on $(1 - r_x)$ and V_x respectively, deliberately treat target-band reliability and between-seed dispersion as secondary penalties. These terms still increase the loss when a model is unreliable across seeds or overly sensitive to stochastic variation, but they should not dominate the loss where the model's central tendency is empirically plausible. Although 0.5 and 0.25 are the natural choices for these weights, these values are still somewhat arbitrary and we should ensure this loss metric validation is robust for comparison across a variety of weight choices.

3.2.1 Experimental procedure

To ensure this loss metric is comparatively stable, I conduct a sensitivity analysis over the generic-loss weights W_R and W_V. I aim to test whether the comparison between model versions depends on the particular weightings in Equation 3.2 or whether comparison is robust so long as $1 > W_R > W_V$ holds. For this sensitivity test, I hold the central-distance weight fixed at unit weight and vary the secondary penalties over an ordered grid that preserves the intended priority structure of the loss ($1 > W_R > W_V$). I choose to vary the reliability weight (W_R) over $\{0.25, 0.5, 0.75\}$ and the dispersion/inter-seed variability weight (W_V) over $\{0.125, 0.25, 0.375, 0.5\}$, whilst retaining only the weight combinations where the dispersion/shape penalty remains smaller than the reliability penalty, to ensure $1 > W_R > W_V$. This yields 8 total weight combinations. The current weights, 0.5 and 0.25, are therefore one point in a wider family of loss functions that encode the same qualitative ordering of central accuracy first, reliability second and dispersion/shape third.

For the bounded-low-is-better metric family (**low**), I chose to treat the JSD level component ($\bar{\bar{x}}$ / U_x) as part of the secondary dispersion/variation penalty rather than as a separate central-accuracy term. This choice avoids giving the distributional realism metrics a fixed extra weight outside the sensitivity sweep.

For each point on this grid, I recompute the validation loss for Carro et al.'s 2011-targeted housing market ABM (*the original model*) and an output calibrated version of this model, which I introduced in Section 5.1 (*the optimised model*). I ensure to use the same ten seeds and the same post 500 time step warm up period results for both models. No model simulations are rerun as I only rescale the already recorded loss components, to isolate the comparison to the chosen loss weights.

3.2.2 Results

Comparing total validation loss between the original and optimised model versions is highly stable. Under the default weights (0.5 and 0.25), the ABM composite loss falls from 0.5652 for the original model to 0.5212 for the optimised model, a reduction of 0.0441, or approximately 7.8%. As represented in Figure 3.1, across the tested weight grid, the optimised model has lower aggregate loss than the original model in every case, with the improvement ranging from 5.7% to 9.0%, hence, the optimised model remains preferred to the original model by overall composite loss across all tested weights within a 3.2%pp range. In this Figure, the grey cells indicate where the dispersion weight is not smaller than the reliability weight.

Per-metric loss rankings are also stable. In Figure 3.2, I correlate the individual metric rankings under each tested weighting, compared to the current weighting I use ($W_R = 0.5$, $W_V = 0.25$). We find the Spearman correlations of at least 0.995 for the original model and at least 0.988 for the optimised model. In addition, the same five worst-performing metrics are identical within each model version under every tested weighting, although the optimised model has a minor ordering swap at the highest tested secondary-weight setting ($W_R = 0.75$, $W_V = 0.5$). This suggests that the loss function identifies the same main validation weaknesses and strengths across the tested secondary-weight choices, and that the chosen weights, $W_R = 0.5$ and $W_V = 0.25$, are not responsible for an idiosyncratic ranking of metric-level weaknesses.

At an individual metric level, comparative signs are mostly stable. As presented in Figure B.1, nineteen of the twenty scored metrics have the same improvement or regression sign for the optimised model relative to the original model under every tested weighting. The only unstable case is HPI Mean: it improves under seven of the eight settings, including the current weighting, but regresses slightly under the lightest secondary-weight setting ($r = 0.25$, $d = 0.125$). This also indicates that the current weighting, $W_R = 0.5$ and $W_V = 0.25$, lies within the stable central region of the tested grid as it agrees with the majority of weighting choices.

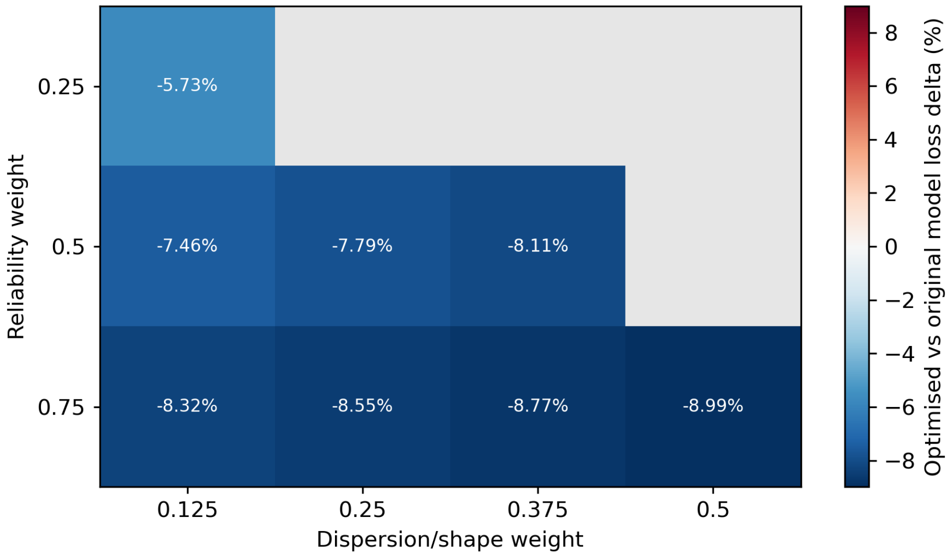

Figure 3.1: Overall loss stability of validation metric losses under alternative loss-weight settings.

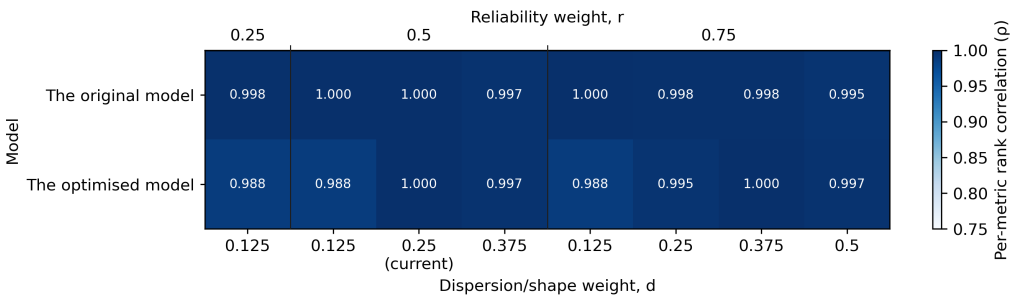

Figure 3.2: Spearman rank stability of validation metric losses under alternative loss-weight settings.

Overall, the sensitivity analysis supports the intended interpretation of the validation loss as a comparative validation criterion. My loss consistently ranks the optimised model ahead of the original model, and it identifies a stable set of metric-level strengths and weaknesses despite variation in the non empirically estimated secondary penalty weights.

3.3 Validating the original UK housing model

Using this new stable per-metric loss function, in Table 3.2 I yield the validation results for Carro et al.'s 2011 target-year calibrated housing market ABM [10].

The validation in Table 3.2 provides evidence that the model reproduces several important aggregate and distributional features of the UK housing market. However, the metric-level losses show that important discrepancies still remain in the model, particularly for leverage, price-income affordability, transaction volumes, and buy-to-let activity.

Finally, it is important to mention that the absolute value of the composite loss (0.565) has no stand-alone statistical meaning. I designed the loss such that its value is most informative when used comparatively, either by using the total composite loss as an aggregate indicator to compare alternative calibrations or model versions, or at the per-metric level to identify which empirical targets are driving improvements or deterioration. The main contribution of this metric is methodological in that I convert a heterogeneous set of validation outcomes into a common, dimensionless and diagnostically useful framework, combining central fit, between-seed robustness and target-band reliability. In Chapter 5, I will use this loss metric extensively to evaluate whether recalibration improves empirical fit under the proposed validation loss relative to the original 2011 calibration.

Metric x	**Unit**	**Target range** L_x, U_x	**Mean value** $\bar{\hat{x}}$	**IQR** $\mathrm{IQR}_{\hat{x}_i}$	**Loss** $\mathcal{L}_x$
Mortgage Approvals	count/month	42,900 to 52,900	53,200	737	0.259
Housing Transactions	count/month	68,700 to 77,200	100,000	1,480	0.767
Advances to FTB	count/month	15,300 to 16,900	23,100	141	0.814
Advances to Home Movers	count/month	25,100 to 27,700	18,900	377	0.790
Advances to BTL	count/month	6,650 to 7,350	11,300	239	0.934
HH DTI	%	162 to 164	99.9	1.21	0.985
House PIR	ratio	4.23 to 4.27	6.40	0.107	0.909
House Price Growth	%	-0.945 to 0.170	0.0497	0.0433	0.0114
HPI Mean	index	0.943 to 1.04	1.05	0.255	0.569
HPI Std	index	0.263 to 0.291	0.224	0.0532	0.618
HPI Cycle Period	months	163 to 180	147	42.8	0.421
RPI Mean	index	1.00 to 1.02	1.02	0.0491	0.462
HH Owning Share	%	65.5 to 66.5	62.8	0.262	0.528
HH Private Renting Share	%	16.5 to 17.5	20.4	0.257	0.530
OO DTI	%	98.6 to 99.8	71.7	0.525	0.821
Rental Yield	%	5.80 to 6.41	4.34	0.0480	0.792
Interest Rate Spread	pp	2.40 to 2.63	3.00	0.00446	0.642
Income Realism	JSD	0 to 0.12	0.0404	0.00106	0.0865
Housing Wealth Realism	JSD	0 to 0.12	0.0784	0.00436	0.172
Financial Wealth Realism	JSD	0 to 0.12	0.0916	0.00139	0.194
Total Composite Loss	-	-	-	-	0.565

Table 3.2: Validation losses for the original 2011 target-year ABM

Chapter 4

Optimising computational performance through caching and parallelisation

Computational performance is a practical constraint on the use and accessibility of agent-based models (ABMs), because calibration, validation, sensitivity analysis, robustness checks and policy experiments all require the model to be executed repeatedly across many different candidate model configurations, starting seeds and experimental scenarios. Moreover, ABMs, including this housing-market ABM, have individual simulation runs with many heterogeneous agents, stochastic decisions and market interactions, so inefficient code paths can quickly become a limiting factor on the scale of experiments that can be conducted. Existing ABM literature identifies computational cost as a major challenge for calibration and analysis, since robust calibration and sensitivity analysis often require many repeated model evaluations, while individual ABM runs can themselves require substantial CPU time [7, p. 367].

In this chapter I present two performance contributions on Carro et al.'s ABM [10]. Firstly, I introduce a cache optimisation which reduces the wall-clock runtime of an individual simulation by 7.6%. Secondly, I introduce the support for the parallel execution support for running independent simulations concurrently, which leads to a further 4.85× increase in throughput relative to a sequential execution on a lightweight model run benchmark[1] and a 5.8× throughput increase on a more realistic benchmark[2]. Combining these optimisations provides a 6.1× throughput improvement relative to the original sequential model. This significantly improves the practicality of using the model for repeated experiments whilst preserving exact model behaviour.

4.1 Reducing per-run runtime via a landlord rental income cache optimisation

Per simulation runtime is a major constraint on the scale of experiments that can be conducted in an ABM, as even small inefficiencies in each run are amplified across many candidate model configurations and seeds per configuration. In this section I focus on reducing the wall-clock time of a single ABM simulation run, whilst preserving exact model behaviour, via a cache optimisation for landlord rental income.

4.1.1 Landlord rental income cache optimisation

The optimisation I made targeted the unnecessarily repeated computation of landlord household's rental income. In the original model implementation, each call to our optimisation's target function, called `getMonthlyGrossRentalIncome()`, recomputes the landlord's monthly rental

[1] On batches of model runs, each with a 5,000 agent population.

[2] On batches of model runs, each with a 10,000 agent population.

income by iterating over all current rental contracts the household has let out as a landlord and then sums each contract's next payments. For a household h at simulation time step t, with current rental agreements $\mathcal{R}_h(t)$, the method therefore evaluates:

$$I_h^{\text{rent}}(t) = \sum_{r \in \mathcal{R}_h(t)} \texttt{nextPayment}(r). \tag{4.1}$$

This value is a deterministic function of the landlord's current rental-contract state and therefore repeatedly recomputing the same sum introduces avoidable computational work without changing the model's economic logic.

The optimisation replaces this repeated full scan using a cache. Each household stores a cached value for monthly gross rental income, together with a dirty flag indicating whether the rental-contract state has changed since the value was last computed. When the target function is called, the cached value is returned immediately if the dirty flag is false. If the dirty flag is true, the method recomputes the same sum over the landlord's current rental contracts, stores the result, and marks the cache as clean. This ensures that the cache is only an implementation optimisation by changing when the value is computed, but not the value returned by the model. The cache is invalidated whenever the rental-contract state can affect the value of $\texttt{nextPayment}(r)$, for example, when a landlord receives a new rental contract or when a letting agreement ends. The implementation also avoids maintaining the cached value through incremental arithmetic updates.

I hypothesised this optimisation would decrease runtime because rental income is queried much more frequently than rental contracts actually change. In the $20,000$ household benchmark with $2,000$ time steps, the household loop alone represents approximately 40 million household-step visits. During each of these visits, rental income is queried repeatedly through gross income, annual gross income, net income, disposable income, consumption, saving-rate calculations and household statistics recording. In contrast, the set of rental contracts changes only when a small number of rental-market lifecycle event occurs. The cache therefore turns most calls from a scan over the landlord's rental-contract map into a constant-time dirty-flag check and cached-value return.

4.1.2 Complexity analysis of the caching optimisation

Let $n_h^{\mathcal{R}}(t) = |\mathcal{R}_h(t)|$ be the number of active rental contracts held by landlord household h at time step t. In the original ABM implementation, each call to `getMonthlyGrossRentalIncome()` recomputes rental income by scanning all of these contracts so, as presented in Equation 4.1, the cost per query is $\Theta\big(n_h^{\mathcal{R}}(t)\big)$.

With the cache optimisation, a query returns the stored value whenever the cache is clean so a cache hit has a constant time cost of $\Theta(1)$. If the cache is dirty, the method recomputes the same rental-income sum as before, so a cache miss still costs $\Theta\big(n_h^{\mathcal{R}}(t)\big)$.

This optimisation does not reduce the worst-case cost of a dirty query, but it changes the common case from a linear scan over the landlord's rental contracts to a constant-time lookup. Since rental income is queried repeatedly while rental-contract changes occur relatively infrequently, most calls are expected to be cache hits, making the expected per-query cost close to $\Theta(1)$.

4.1.3 Benchmarking protocol

To evaluate whether an optimisation improves per-run execution time, I use a two-stage protocol to separate model correctness from runtime measurement. Firstly, I run a smaller correctness check. The ABM is run with a small number of households $(5,000)$ for $2,000$ time steps. I then directly compare all byte-for-byte exactness of all 25 of the model's outputs compared to pre-existing outputs from the model without any code changes. This ensures any optimisations do not alter any model behaviour. This is also repeated three times to ensure execution is deterministic.

Only if this correctness check is passed, we measure runtime using a larger benchmark with 20,000 households, 2,000 simulation steps and ten repeated simulations runs each for the default model and cache on version, with each run pinned to a single core and ran sequentially on the same seed to benchmark and on the same stochastic workload. The benchmark size was chosen so that model computation dominates runtime over fixed overheads such as output generation, while still being cheap enough to run repeatedly in a realistic time frame. Importantly, it also provides enough timing resolution to detect small but practically useful improvements.

On a baseline benchmark, I found the mean model per-run runtime to be $\bar{t} = 43.469086 \pm 0.748709$s across $n = 10$ runs (uncertainty uses 95% CI half-width). Assuming similar variance for the optimised version, the standard error of the difference between two independent ten-run mean runtimes is:

$$SE_{\Delta} = \sqrt{\frac{2s^2}{n}} \approx 0.468\text{s}.$$

Using a 95% confidence interval (CI) with 18 degrees of freedom yields $t \approx 2.10$, so the approximate uncertainty around the measured runtime difference is 0.983s and as a percentage of the baseline runtime, this is approximately 2.26%. This means that, under similar run-to-run variance, the benchmark should be able to distinguish a speed-up of around 2.5% from random timing noise. A 2.5% improvement corresponds to approximately 1.09s, which is slightly larger than the estimated 95% uncertainty of 0.983s.

This threshold is useful because I expected any local code optimisations I make in this ABM to produce incremental speed-ups rather than order-of-magnitude improvements. I therefore chose this benchmark size to be sensitive enough to detect small per-run runtime improvements, but not so expensive that it becomes impractical to repeat during development.

4.1.4 Optimisation result

Runs each	Agents	Seeds	Core pinned	Mean speedup	Mean runtime delta %	Mean runtime 95% CI	Median runtime delta %	p95 runtime delta %	Std. dev. delta %
10	20k	1	Yes	1.082x	-7.59%	[4.27%, 10.81%]	-8.72%	-6.46%	+0.76%
400	10k	1-40	No	1.084x	-7.61%	[3.21%, 12.00%]	-6.72%	-7.94%	+0.02%

Table 4.1: Runtime impact of the cache optimisation for independent UK ABM executions.

For this experiment, as presented in row 1 of Table 4.1, the default model had mean wall-clock runtime of 45.27s while the cached implementation had mean wall-clock runtime of 41.83s, a **7.59**% reduction in per-run execution time on our 10 run benchmark, with a 95% confidence interval of 4.20% to 10.98% reduction in runtime.

Distributional effects of the cache optimisation

To analyse the distributional effects of this cache optimisation on runtime, I use Figure 4.1 which plots the runtimes of 400 runs, across 40 different seeds, each with a agent population of 10,000; a realistic policy experimental workload across different stochastic workloads. This data is reused from our final runtime analysis presented in Section 4.3. As evidenced in Figure 4.1 and row two of Table 4.1, I found both with and without the cache optimisation, model runtimes across seeds are unimodal but positively skewed, however, enabling the cache shifts the distribution left, without a major loss in variation. The mean and p95 per-run runtimes fall by 7.6% and 6.72% respectively. The sample standard deviation increases only 0.02% and hence total

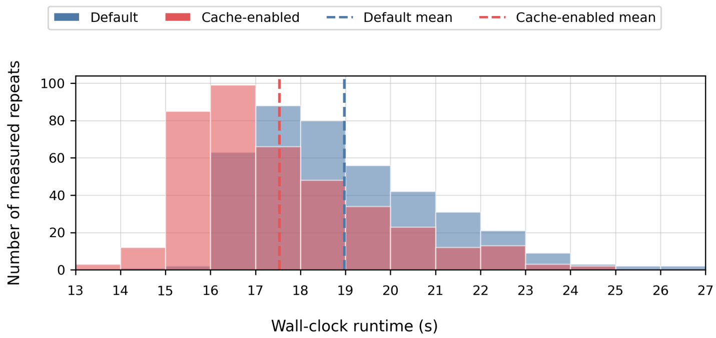

Figure 4.1: Histograms of runtimes of default model executions compared to executions with the rental cache optimisation enabled across 400 runs each using a 10k agent population.

runtime variability is effectively unchanged. Finally, the p90 and p99 decrease proportionally to the decrease in median runtime. Hence, this optimisation acts primarily as a consistent distribution-wide speed-up, lowering typical and high-percentile runtimes without introducing additional variability.

Analysis of cache optimisation hit-rate

To further explain the source of this speed-up, I also ran an instrumented version of the cached model which recorded cache queries, re-computations and invalidations without changing the model logic. This profiling run used 20 seed model runs, each with an agent population of 20,000 for 2,000 time steps. Across these runs, `getMonthlyGrossRentalIncome()` was queried 5.95 billion times, but 99.8% of calls were clean cache hits, so only 0.2% required a dirty recomputation. Even when I restricted attention to landlords with at least one active rental contract, the cache hit rate remained 98.0%. The model would have scanned 1.2 billion rental-contract entries without the cache, whereas the cached implementation scanned only 39.3 million, avoiding 96.7% of the previous contract-scanning work. This confirms rental income is read extremely frequently compared to rental-contract lifecycle changes which means most invalidations are amortised over many subsequent constant-time reads.

Effects of ABM agent population on cache optimisation

To understand how this speed-up is affected by agent population size, I chose to run another experiment, varying agent population size on the model to understand if this effects the speed-up caused by this cache optimisation. For this experiment, I chose to run forty model runs per population size for each the default model and the cache-optimised model. I ensured to pin runs to one core and randomise and interleave runs by population. I then compare the end-to-end execution times between agent population sizes.

Agents	Mean speedup	Mean speedup 95% CI	Mean runtime delta %	Mean runtime 95% CI
5,000	1.044	[0.999, 1.091]	-3.26%	[-7.50%, +0.98%]
10,000	1.064	[1.030, 1.099]	-5.50%	[-8.55%, -2.44%]
20,000	1.066	[1.052, 1.079]	-6.08%	[-7.28%, -4.88%]
50,000	1.076	[1.060, 1.092]	-6.97%	[-8.35%, -5.59%]

Table 4.2: Effect of the cache optimisation whilst varying agent population size.

Figure 4.2 and Table 4.2, show that this cache optimisation provides a consistent runtime improvement across all measured ABM population sizes and also the optimisation gives larger

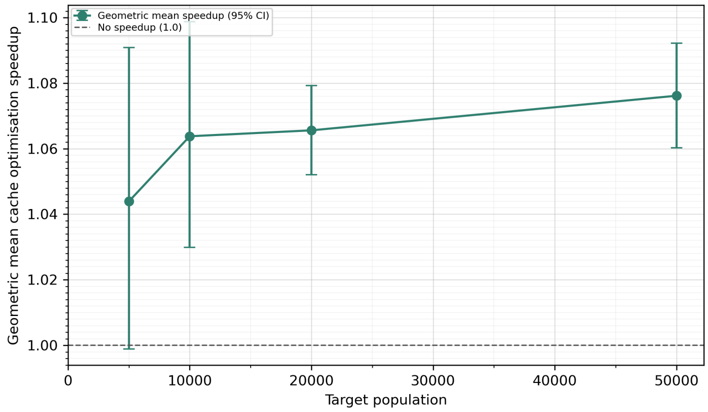

Figure 4.2: Mean speed-up of cache optimisation whilst varying agent population size.

and more reliable benefits as agent population increases. Runtime reduction rises from 3.26% at 5,000 agents to 6.97% at 50,000 agents. This suggests the optimisation becomes more useful when repeated rental-income queries dominate fixed runtime overheads.

4.2 Reducing batch simulation runs through parallelisation

Parallelisation is the process of splitting a computational task into several smaller separate parts which allows multiple processors to work on each of the smaller tasks [27, p. 1], which ideally leads to a reduction in the runtime of a computation. For this ABM, we often require many independent model runs across starting seeds and candidate parameter sets, each of which do not need to communicate with each other whilst they execute.

It is therefore likely advantageous to assign each of these independent runs to a separate processor and then aggregate results afterwards, rather than executing these runs sequentially. I ideally expect this strategy to reduce the wall-clock time required to complete a batch of experiments. However, the benefit of this form of parallelisation cannot be assumed from the independence of the runs alone. Although independent simulations require no model-level communication, they will still likely compete for shared system resources in a computer system including shared cache, memory bandwidth and JVM runtime overhead.

4.2.1 Parallel scaling experiment

In this subsection, I aim to evaluate the extent to which independent ABM runs can be accelerated at a batch level. I aim to quantify how scaling the number of 'workers' affects throughput or the number of independent ABM executions completed per hour. I define a *worker* as a single concurrent execution unit which is responsible for executing one independent ABM run at a time. This follows the general worker-pool model, where workers are assigned tasks, execute them, and then become available for further work [27, p. 370]. In this experiment we implement the worker abstraction at the process level, not the thread level, as each worker needs to launch a separate Java process with a fixed seed and an independent output directory. This isolates each model run, avoids modifying the internal logic of the ABM, and allows the experiment to measure how completed-run throughput changes as additional independent simulations are scheduled concurrently.

For this experiment, I selected the workload as ABM runs with a target population of 5000 households and 2000 time steps per run. I evaluated model run throughput under a variety of worker counts from one to thirty-two workers with each worker-count batch consisting of forty independent seed runs. I also ran three repeated batches per worker count to ensure stability. To isolate the effect of worker count, all runs used the same commit, input snapshot, seed sequence, machine, and output mode. I ran this experiment on my machine with 20 logical processors[3]. I also chose to run a 24 and 32 worker count to explore the effect of worker count surpassing the number of available cores on the machine.

The results in Figure 4.3 and Table 4.3 show that the parallelisation framework substantially increases model-run throughput, but this scaling is sub-linear. A single worker completed a mean of 332.0 model runs per hour and 20 workers completed 1610.5 model runs per hour which corresponds to a 4.85× speed-up.

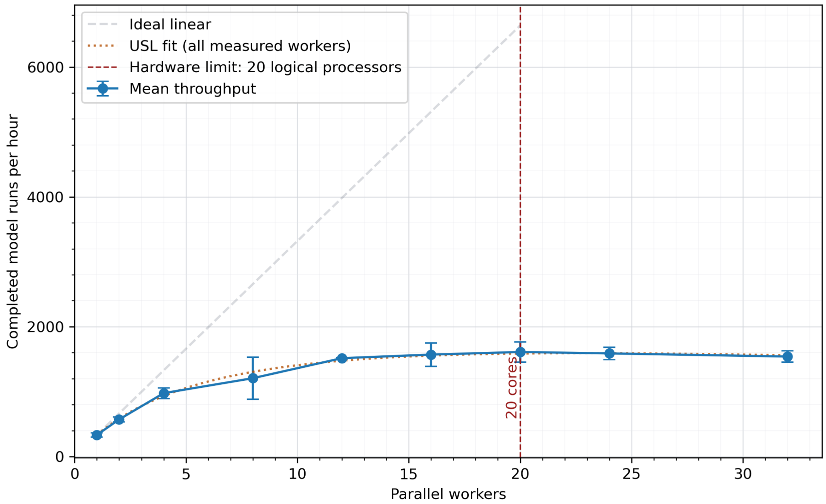

Figure 4.3: Effect of the number of parallel workers on the throughput of independent ABM runs at a 5k population.

Although this is a significant speed-up, as scaling is below twenty times the single-worker throughput, this indicates that model runs are not independent at the hardware and runtime level. The largest gains occur when moving from low worker counts to moderate worker counts. Increasing from 1 to 2 workers adds 239.6 completed runs per hour per additional worker compared to the 332.0 completed runs per hour using 1 worker. Increasing from 2 to 4 workers adds 202.4 completed runs per hour per additional worker. After this point, the marginal benefit declines with the gain from 12 to 16 workers only being 13.7 runs per hour per additional worker, and the gain from 16 to 20 workers only being 10.1 runs. Scaling efficiency also therefore decreases from 0.861 with 2 workers to 0.243 with 20 workers. Therefore, the experiment demonstrates that additional workers remain useful up to the hardware limit, but with sharply diminishing returns.

Beyond the 20-logical-processor hardware limit, throughput regresses with mean throughput falling from 1610.5 runs per hour at 20 workers to 1588.3 at 24 workers and 1540.3 at 32 workers. Both settings therefore have negative marginal gains. This suggests that over-subscription introduces enough scheduling overhead, JVM contention, cache pressure, memory-bandwidth pressure or other shared-resource contention to outweigh any benefit from keeping more processes queued. For this workload and machine, the best observed setting is therefore 20 workers.

Modelling speed-up using the Universal Scalability Law

To characterise the observed sub-linear scaling more formally, I fit these results to the Universal Scalability Law (USL) [28]. The USL naturally models the effects of contention, coherency and over-contention in parallel systems. This is the natural regression for this kind of experiment

[3]We define *logical processors* as the hardware execution contexts exposed by the CPU to the operating system scheduler which are analogous to hardware threads.

Workers (w)	Throughput[a] ($\bar{T}_w$)	95% CI[b]	Speed-up[c] (S_w)	Efficiency[d] (E_w)	Marginal gain[e]
1	332.0	32.5	1.00×	1.000	-
2	571.6	36.4	1.72×	0.861	239.6
4	976.4	81.1	2.94×	0.735	202.4
8	1206.6	324.0	3.63×	0.454	57.6
12	1515.3	11.9	4.56×	0.380	77.2
16	1570.1	179.1	4.73×	0.296	13.7
20	1610.5	154.0	4.85×	0.243	10.1
24[f]	1588.3	96.9	4.78×	0.199	-5.54
32[f]	1540.3	86.9	4.64×	0.145	-6.00

[a] Throughput is mean completed model runs per hour.
[b] CI is the 95% confidence interval half-width.
[c] Speed-up is the ratio of mean throughput at w workers to mean throughput with one worker: $S_w = \bar{T}_w/\bar{T}_1$
[d] Efficiency is parallel scaling efficiency: $E_w = S_w/w$
[e] Marginal gain is the change in throughput per additional worker versus the previous worker setting.
[f] The 24- and 32-worker settings exceed the 20 logical processors available on the test machine, so they represent over subscription beyond the hardware limit rather than additional physical capacity.

Table 4.3: Parallel scaling throughput for independent UK ABM executions.

as the it expresses throughput as a function of the number of concurrent workers and allows departures from the ideal linear scaling to be interpreted in terms of shared-resource contention and coordination overhead. I found regressing using this gives a high of $R^2 = 0.943$. The regression gives the USL equation:

$$\hat{T}(w) = \frac{\bar{T}_1 w}{1 + \alpha(w-1) + \beta w(w-1)}, \ \bar{T}_1 = 332.0, \ \alpha = 0.1347, \ \beta = 0.00165 \tag{4.2}$$

In Equation 4.2, α is the contention coefficient and represents the degree to which introducing additional workers introduce serialising delays through competition for shared resources, such as CPU time, memory bandwidth, disk access, or process-management overhead. The fitted value of $\alpha = 0.1347$ therefore indicates that a significant proportion of the lost scalability is attributable to contention. The parameter β is the coherency coefficient and captures coordination costs between workers. $\beta = 0.00165$ and also β is far smaller than α which indicates that worker scalability is more limited by contention for resources than worker communication. This aligns with my expectations as there is no inter-process communication (IPC) when executing model runs in this parallelisation framework. β therefore likely captures higher-order shared-resource effects and over-subscription penalties.

Analysis of the effect of ABM population on parallelisation scaling

To test if this sub-linear scaling is stable across model workload sizes, I also chose to run an experiment where I vary the target agent population of the model and analyse the effect this has on relative model run throughput due to parallelising across worker counts. I chose to run the same experimental methodology as for the absolute speed-up experiment using 2000 time-steps per run and three batches of forty model runs per worker count, but I also vary agent population between 5,000, 10,000 and 20,000 population sizes. I instead plot relative throughput to running the model with one worker at each population size for effective comparison between population sizes. To avoid the effect of thermal drift influencing the order of results, I also chose to randomise the order of both the population and worker counts of these batches of runs.

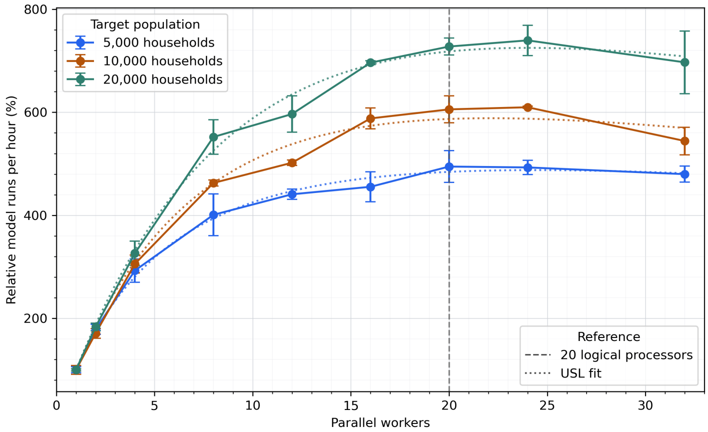

Figure 4.4: Effect of the number of parallel workers on the relative throughput of independent ABM runs across 5k, 10k, 20k population sizes

Agent population	**20 worker relative throughput**[a]	**USL R^2 fit**	**USL α**	**USL β**
5,000	4.946x	0.982	0.1357	0.0014
10,000	6.057x	0.985	0.0895	0.0019
20,000	7.276x	0.986	0.0615	0.0016

[a] 20 worker throughput is mean completed model runs per hour at twenty workers relative to the mean completed model runs per hour at one worker for this respective population size.

Table 4.4: Analysis of parallel scaling throughput relative usign multiple workers for independent UK ABM executions.

As presented in Figure 4.4 and Table 4.4, I find the parallelisation framework becomes more effective as population size increases. At the twenty worker hardware limit, relative throughput rises from 4.95x at 5,000 agents to 6.06x and 7.28x at 10,000 and 20,000 agents respectively. This shows that the results from our original 5,000 population scaling experiment, found in Table 4.3 and Figure 4.3, are actually a conservative speed-up from this parallelisation due to a small run size. Larger populations seem to make better use of the worker pool, which is likely caused by larger ABM runs amortising fixed overheads more effectively, so a bigger proportion of wall-clock time is spent in useful computation.

Fitting a USL regression across each population size yields strong and consistent fits, with R^2 fitting ranging from 0.982 to 0.986. I found the contention coefficient (α) falls from 0.1257 at 5,000 households to 0.0615 at 20,000 households. It is likely that smaller 5k workload seems to spend a larger fraction of runtime in fixed parallelisation and runtime overheads. All three curves still plateau around the 20-logical processor limit with over-subscription not providing any consistent benefit.

In summary, I found this parallelisation framework is more effective for larger, more realistic experiment batches, leading to up to a 7.28x throughput increase for larger 20,000 population batches at 20-cores, with smaller runs scaling less due to fixed overheads dominating.

Summary

Through this parallelisation framework and this experiment, I demonstrate that this process-level worker framework can lead to a significant reduction of runtime, leading to a 4.85x to 7.27x increase in batched ABM run throughput, or equivalently a 79.4% to 86.2% reduction in runtime for batched model runs as we vary ABM agent population from a 5,000 to a 20,000 population respectively. I also theorise this speed-up is limited largely due to resource contention which is more pronounced at lower population sizes. Regardless, this framework materially improves the feasibility of large simulation batches due to these significant performance gains, even running on my local machine.

4.3 Analysing model combined speed-up

To estimate the final combined speed-up of the model under both the cache optimisation and the 20-core parallelisation framework, I chose to run a blocked 2x2 factorial benchmark, where each cell ran the same fixed workload under four conditions: no cache with one worker (A), cache enabled with one worker (B), no cache with 20 workers (C) and finally cache enabled with 20 workers (D). I chose a workload of 10 blocks for each of the 4 cells, with each individual block is a more realistic batch workload of 40 seed runs at a 10,000 model population and 2000 time steps. This means in total 1,600 runs were executed for this benchmark. To reduce temporal bias, the ratio was computed within each block on the log scale as $\log D_i - \log A_i$, then averaged across the 10 blocks. Speed-up was reported as the exponentiated mean log ratio a 95% confidence interval calculated using the standard error of the paired block log ratios and a t critical value with 9 degrees of freedom.

Through running this experiment on a more larger realistic workload than in our original 5,000 population throughput experiment in Section 4.2), I obtain the speed-up results presented in Table 4.5. I find that the combined effects of the cache optimisation and the parallelisation framework leads to a total speed-up of 6.1× or alternatively a 83.7% decrease in total runtime with a cache parallelisation interaction of 0.971×. I observe the cache optimisation per-run runtime decrease of 7.7% in this experiment, which is very close to our 7.6% runtime reduction result obtained in Subsection 4.1.4. I also, again, observe a significantly higher speed-up in this experiment of 5.8× with a 10,000 household population compared to the original parallelisation framework experiment with a speed-up of 4.9× with 5,000 household population. This is, as

Cell	Cache	Workers	Speed-up	95% CI	Runtime Decrease	95% CI
A	Off	1	1.000×	-	0.0%	-
B	On	1	1.084×	[1.033×, 1.136×]	7.7%	[3.2%, 12.0%]
C	Off	20	5.833×	[5.569×, 6.110×]	82.9%	[82.0%, 83.6%]
D	On	20	6.136×	[5.809×, 6.481×]	83.7%	[82.8%, 84.6%]

Table 4.5: Blocked factorial benchmark results for cache and worker parallelism effects. Speed-up and runtime decrease are reported relative to the no-cache single-worker baseline.

previously mentioned, likely due to the larger benchmark leading to a larger speed-up as a smaller fraction of wall-clock time is spent in fixed parallelisation overhead and instead is spent directly in model computation.

Chapter 5

Improving and updating ABM calibration

5.1 Improving output calibration using TuRBO

5.1.1 Output calibrated parameters

Certain input parameters in the housing market ABM represent intrinsic factors in behaviour in the housing market. For these parameters, it is not possible to empirically calculate and estimate from survey data nor assume plausible values; we instead need to calibrate these parameters directly from the simulated model outputs, rather than empirically estimating using available data [10, C.3]. We will therefore refer to these parameters as output calibrated parameters. Output-based calibration of ABMs is an active methodological challenge, as it is often difficult to fit ABMs to empirical data due to high computational cost, stochastic uncertainty, and limited consensus on the most robust estimation methods [8, Sec. 1].

In Carro et al.'s original 2011-calibrated housing market, there are five of these output-calibrated parameters [10, C.3] which are:

1. **Buy-to-let probability adjustment**: This parameter scales the probability that a household is given the 'buy-to-let flag', which when given to a household allows the household to become a buy-to-let investor. This flag therefore accounts for the fact that the model must represent not only households that are currently BTL investors, but also households with a tendency to invest whom may be prevented from doing so by constrained income or wealth.

2. **Sensitivity for BTL buy and sell decisions**: This parameter controls how strongly buy-to-let investors react to expected investment yield when deciding whether to buy a new rental property or sell an existing one. A higher value makes BTL behaviour more responsive to calculated profitability, while a lower value makes these decisions more random.

3. **Market-average price decay**: This parameter represents how households form their view of house prices between the overall house market movements and the prices in the specific quality segment they are considering. A higher weight on the segment means households form their view more from the house prices of properties with a similar quality to the ones they are viewing, whilst a higher weight on the market means they rely more on the general house price index.

4, 5. **Psychological cost of renting** and **Sensitivity between rent or purchase**: In this model, households treat renting as an additional perceived cost compared to owning, so households may still prefer to buy even when the direct financial cost of renting is similar.

The psychological cost of renting parameters captures the preference households have for owning rather than renting, with a higher value making renting less attractive relative to ownership. Sensitivity between rent or purchase determines how strongly households respond to the differences between the perceived cost of renting and the cost of buying, with higher values making the choice more deterministic and lower values allowing greater variation in household decisions.

5.1.2 Current output calibration methodology

In this 2011 target-year calibrated model, which we will refer to as the original model, output calibration was done via a grid-based simulated method of moments (SMM). Carro et al. select five to eleven candidate parameters for each of the five output calibrated parameters in the model. The model is then ran for ten simulations, each for 3500 time steps per simulation, for every parameter combination out of a total 26,730 combinations. Carro et al. then select the parameter combination which leads to the best criterion function value, based on the average squared percentage errors between simulated and empirical moments [10, B.2].

This output calibration strategy is effective because it provides a systematic way to calibrate these output calibrated parameters. By selecting the parameter combination that best matches several empirical moments simultaneously, Carro et al. ensure that the chosen values are judged by their contribution to the overall realism of the simulated housing market, rather than by whether each parameter appears individually plausible. However, I believe this calibration approach could be improved in two important aspects. Firstly, this method is computationally expensive, even with relatively few candidate values for each parameter. The full grid search requires $267,300$ full ABM simulation runs, and this cost would grow rapidly if more parameters or finer candidate grids were introduced. Second, the resolution of the search is constrained by the predefined grid of candidate values, meaning that the selected combination is only the best among the values tested, rather than necessarily being close to the true optimum in the continuous parameter space. These limitations motivate the use of more efficient calibration methods which can explore the parameter space more flexibly while requiring fewer full model evaluations.

5.1.3 Trust Region Bayesian Optimisation (TuRBO)

Stochastic agent-based models, such as this UK housing market ABM, are often treated as black-box (or unknown) functions during calibration, because their mapping between input parameters and model output is generally evaluated by running a simulation of the model as the model is generally not available in a closed form [8, Sec. 1]. Bayesian optimisation (BO) is a method for finding the optimal input parameters for a given black box function and is well suited to this task as it uses previous input and output combinations to build a stand-in model, called the surrogate model, instead of the true unknown function. BO uses this model to predict and find new candidate parameters that will likely improve the calibration loss [29, pp. 148-149].

Trust Region Bayesian Optimisation (TuRBO) was introduced in 2019 and extends BO by using a local surrogate model instead of one global surrogate model. This model is only calculated in its own small trusted region around an area of promising input parameters. This local surrogate model's trusted region can adaptively shrink and expand based on whether experimented runs improve or worsen solutions. This allows TuRBO to concentrate candidate expensive simulation runs in promising areas [11]. This is attractive for output-calibrating an ABM because each objective-function evaluation requires many full stochastic simulations, and the resulting loss surface may be noisy, non-linear and uneven across the parameter space. TuRBO starts off with an exploratory period, where a small set of parameter configurations are selected relatively evenly across the search space to explore the space before Bayesian optimisation begins. I use a sampling technique called Sobol sampling to explore the parameter space which is inspired by BoTorch's TuRBO implementation to generate the initial points for a BO loop [30, Generate initial points].

I took inspiration to use TuRBO from Bai et al.'s calibration of a financial-market simulator by minimising a distributional distance between simulated and target output series, where they show TuRBO outperforms standard Bayesian optimisation [31].

5.1.4 Recalibrating the ABM using TuRBO

For this experiment, I use TuRBO-1, the single-trust-region variant of TuRBO to refine the selection of output calibrated parameters in this housing market ABM. The single trust region variant is appropriate for this ABM calibration problem because the number of output-calibrated parameters is relatively small and each model evaluation is computationally expensive. Although TuRBO-m can improve global exploration by maintaining multiple trust regions in parallel, this requires more evaluations to initialise and more evaluations to update several local surrogate models [11]. TuRBO-1 provides a simpler and more evaluation-efficient choice for this calibration setting, while still retaining the main computational advantage of TuRBO: concentrating model evaluations within an adaptive local region of the parameter space.

I performed TuRBO on the original 2011 target-year housing market with the target of optimising the five output-calibrated parameters defined in Subsection 5.1.1. The optimisation target was to reduce the validation loss we introduced in Section 3.1, whilst also not significantly regressing the amplitude or frequency of house price cycles (HPI standard deviation and cycle period). TuRBO was run with the parameters illustrated in Table 5.1.

Parameter	**Symbol/Role**	**Value**
Candidate replications	Stochastic runs per candidate	10
Simulation Validation Window	Metric Aggregation Window[a]	500-3500
Target population	Household agents simulated	10,000
Optimizer variant	m, trust regions	1
Search dimension	d, parameters optimised	5
Initial observations	Scrambled Sobol points	20
Batch size	q, aligned with 20 workers	2 candidates per iteration
Initial trust-region length	L_{init}	0.8
Minimum trust-region length	L_{min}	$0.0078125 = 0.5^7$
Maximum trust-region length	L_{max}	1.6
Success tolerance	τ_{succ}, batches to expand trust region	10
Failure tolerance	τ_{fail}, batches to shink trust region	3
Observation-noise floor	Minimum $\hat{\sigma}^2$	10^{-6}

[a] Validation loss is calculated over simulated time steps $500 \leq t < 3500$, which excludes an initial warm-up period of 500 time steps.

Table 5.1: TuRBO-1 optimisation configuration summary

5.1.5 Optimisation results

After 1,860 simulation runs across 186 evaluated parameter candidates, we produced a TuRBO optimised version of the 2011 target-year housing market ABM, with parameter modifications defined in Table 5.2.

After running the TuRBO optimisation, I then compute the validation loss of the newly optimised version using our validation function described in Section 3.1 and compare to our original model validity from Section 3.3. The full validation loss comparison statistics can be found in Appendix Table C.1.

From this table, I find TuRBO optimisation leads to several reductions in validation loss:

Parameter	Original Value	Optimised Value	Delta	Delta %
Psychological cost of renting	0.40	0.40	0.00	0.00%
Sensitivity between rent or purchase	0.00100	0.00130	+0.0003	+30.0%
Buy-to-let probability adjustment	1.760	1.825	+0.065	+3.69%
Sensitivity for BTL buy and sell decisions	100	130	+30	+30.0%
Market-average price decay	0.500	0.540	+0.040	+8.00%

Table 5.2: TuRBO optimised output calibrated parameters from the 2011 target-year

- Overall composite validation loss decreases by 7.79%.
- The HPI cycle-period and HPI-standard-deviation losses decrease by 76.3% and 16.4%.
- Model mortgage approval rate validation loss improves by 99.0%, which shows varying these five output calibration parameters significantly influences the validity of this target metric.

5.1.6 Comparing TuRBO to the original model calibration

Additionally, I chose to experiment to answer how much better TuRBO is compared to using Carro et al.'s original grid method in terms of calibration accuracy and convergence speed, if we recalibrated these five parameters from scratch using my new validation loss for both methods. For the original method, I ran the optimisation on the discrete grid of values originally used to calibrate these output calibrated parameters [10, p. 68]. For TuRBO, I ran the optimisation on the continuous interval corresponding to the range of each parameter in the original grid. Per set of candidate parameters, I ran the model across 10 stochastic seeds, with a 10,000 agent population and 3,500 time step runs, as also done during Carro et al.'s calibration.

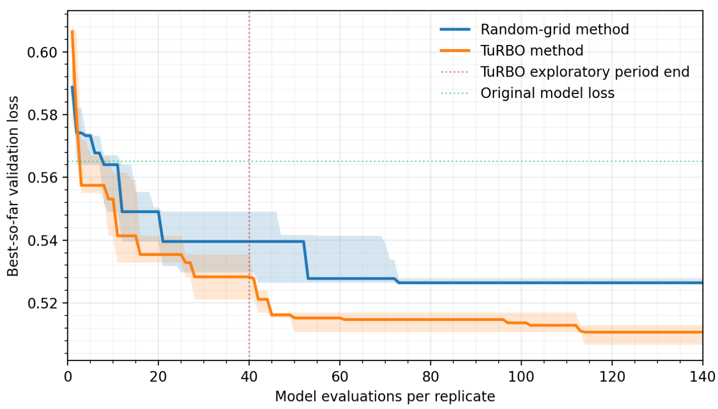

Figure 5.1: Best-so-far validation loss convergence between the random-grid and TuRBO methods.

To ensure a fair comparison of convergence between the original method and TuRBO optimisations, the original method candidate parameters are selected randomly across the grid of potential combinations. Moreover, I ran five independent replicates of each of the two methods. Again, each candidate parameter set was evaluated over ten ABM simulation seeds, and I plotted the median best-so-far composite loss across the five replicates for each method. I ran the methods to 140 candidate evaluations each, as this already amounts to 14,000 model runs. By replicating the original output-based calibration of this model and showing best-so-far

Evaluation	SMM loss[a]± SD	TuRBO loss[a]± SD	TuRBO loss delta (%)[b]	TuRBO advantage[c]
40	0.5390 ± 0.0109	0.5275 ± 0.0122	-2.1%	6.0%
60	0.5327 ± 0.0131	0.5147 ± 0.0066	-3.4%	9.3%
80	0.5262 ± 0.0058	0.5146 ± 0.0066	-2.2%	6.0%
100	0.5262 ± 0.0058	0.5144 ± 0.0066	-2.3%	6.1%
120	0.5262 ± 0.0058	0.5121 ± 0.0071	-2.7%	7.3%
140	0.5262 ± 0.0058	0.5115 ± 0.0059	-2.8%	7.6%

[a] Loss is the best-so-far validation loss across the five replications.
[b] Loss delta is the difference in mean best-so-far loss across replications.
[b] Advantage is defined as the loss delta as a share of the full observed candidate-loss range across all evaluated parameter settings.

Table 5.3: Comparison of best-so-far model calibrations, based on validation loss, using both the SMM and TuRBO methods across five replications each

convergence, we can directly quantify the accuracy and computational benefit of using TuRBO for our refined calibration compared to reusing the grid-based method. Although Carro et al. likely ran the full grid in a fixed order, I aimed to generalise the method to a random-grid to directly compare methodology. For this experiment I chose TuRBOs exploratory period to end after iteration forty. By randomising the order of the original grid, this experiment compares TuRBO against an anytime random-prefix version of Carro et al.'s grid-search calibration. A full exhaustive grid comparison would require evaluating all 26,730 grid parameter sets.

I find from this experiment, that for the same evaluation budget, TuRBO finds better parameter sets than Carro et al.'s grid based method. As presented in Figure 5.1 and Table 5.3, TuRBO outperforms the random-grid method in median best-so-far validation loss at every evaluation after the third evaluation, and from evaluation 45 onwards, all five TuRBO replicates remain below their paired random-grid replicates, only five iterations after the TuRBO parameter-space exploratory period ends. I chose to report both the difference in loss of TuRBO compared to the original grid method at different evaluations, but also an 'TuRBO advantage' metric, which represents how much TuRBO improves loss relative to the variation in loss actually observed across the explored calibration space. This is useful since since this validation loss is unitless composite metric and this allows us to quantify the true advantage of TuRBO in this observed parameter space of these output-calibrated parameters. At the end of the 140 evaluations, on average, TuRBO has a 7.6% advantage over the grid-based method. Furthermore, due to TuRBOs exploration across the continuous parameter space, TuRBO produced a more precise output-calibrated parameter set than the values in the random grid, which is also the case with the TuRBO calibration produced in Table 5.2.

5.1.7 Summary

Beyond the direct improvement in validation loss of 7.79%, TuRBO represents a methodological improvement in how the output-calibrated parameter space is searched and addresses the limitations discussed in Subsection 5.1.2. As previously discussed, Carro et al.'s SMM random-grid calibration evaluates 26,730 parameter combinations, however the TuRBO calibration instead evaluates only 186 candidate parameter vectors. Although TuRBO started from a valid calibration and can likely converge to the trust region faster, this reduction of 26,544 evaluated combinations, corresponds to a 99.3% reduction in candidate evaluations which is computationally significant. Through the performance improvements discussed in Chapter 4 combined with the efficiency of TuRBO, I was able to run this optimisation parallelised on my 20-core local machine in about five hours. I estimate without these model performance and algorithm efficiency improvements, the original grid with 26,730 candidates or 267,300 model runs would instead take about 6.1 months[1] on my same local machine. Additionally, TuRBO also improves the effective resolution of calibration. TuRBO treats the parameters as continuous decision variables rather than a small set of predefined candidate values. This improvement is evidenced by four out of five of the new optimised output calibrated parameters either being in-between subsequent grid values or falling outside the range of the grid completely. In summary, we obtained an output calibration which is more accurate, precise and robust than the original calibration whilst reducing the required computation of executing this calibration compared to the original calibration by 99.3%.

5.2 Recalibrating the UK Housing ABM to 2024

Since the original ABM's target calibration year of 2011, there has been significant shifts in the UK housing market and the policies affecting it. The restriction of mortgage interest relief for individual landlords and introducing the additional-property stamp duty surcharges in 2015/16 led to the share of new buy-to-let mortgage advances falling from 25% in 2015/16 to 12% by 2024, while the share of new first-time buyers' mortgage advances has risen from 23% in 2007 to 41% by 2024 [22]. Additionally, data from the Office for National Statistics (ONS) shows the ratio of median house price compared to median earnings (average housing affordability), rose in England from 6.8 in 2011 to 7.7 in 2024 [32]. Recalibrating the model from the 2011 pre-policy baseline to a more recent period would ideally reflect this weaker buy-to-let investment, stronger first-time-buyer participation, and greater affordability pressure in the UK housing market.

For this project I chose to recalibrate this ABM to the target-year of 2024 to reflect this shift. In 2023, the housing market was still absorbing the mortgage-rate shock associated with tighter monetary policy. There were high borrowing costs and affordability constraints which reduced housing demand, with mortgage approvals and transactions weakening [33]. 2025 was also not suitable as a calibration target because market activity was affected by the expiry of the temporary Stamp Duty Land Tax threshold increase, which brought transaction spike into March 2025 [34]. By contrast, 2024 captures the market after the initial rate shock had been absorbed but before this tax-timing distortion. It also displays signs of market normalisation with annual UK house-price growth reaching 4.6% in December 2024 [35], housing affordability improving in 289 of 318 of the local authorities in England and Wales [36], and net mortgage approvals for house purchase recovering to 66,500 in December 2024 [37]. 2024 therefore provides a recent but comparatively undistorted benchmark year for the UK housing market, whilst still displaying the shift in housing market.

In this ABM, there are 75 unique parameters to calibrate. Some must be empirically estimated from data sources, some are postulated and some are output calibrated (which we discuss in Section 5.1) [10, App. B]. In this section I detail the effort I made to recalibrate this ABM as far as possible to the target-year of 2024.

5.2.1 Calibration strategy

To ensure that recalibration from a 2011 targeted model to a 2024 targeted model is accurate, reliable and robust I chose to employ a variety of methods. Firstly, I chose to recalibrate this model parameter by parameter and storing incremental 'versions' of the ABM parameter set as each parameter is updated. At each of these model parameter versions, I validated the model against the 2024 target-year validation loss metric introduced in Section 3. The advantage of this methodology is that I can quantify the exact effect updating each parameter has on overall ABM loss compared to the 2024 target-year validation loss metric and I can also see how updating a

[1] Assuming we ran these 267,300 runs without our 6.136× speed-up obtained in Section 4.3 through parallelisation and the cache optimisation.

singular parameter affects the validity of each of the twenty validation indicators introduced in Chapter 3.

To recalibrate a parameter, I employ a systematic approach. Firstly, using the original data source used to compute the parameter in Carro et al.'s model, I attempt to reproduce the 2011 target-year calibrated parameter's value. Doing this before every calibration helped me to understand precisely and programmatically how parameters were calculated originally in this model. Per-parameter reproduction is documented in the calibration section of my ABM website/application [38] introduced in Chapter 6. Secondly, I evaluate the methodology employed to compute this parameter based on my own knowledge and identify improvements. As an example, several parameters derived from survey-based micro-data were originally estimated without applying survey weights. Applying these weights is a methodological improvement because it adjusts for survey design, so the resulting parameter estimates better represent the target household population rather than only the sampled respondents. Thirdly, I compute a new parameter value using an equivalent data source from the 2024 target-year, ensuring to include the improvements in methodology I identify after reproducing the 2011 target-year value. For most of the recalibrated parameters it was sufficient to reuse the same data source collected in a more modern time period. For some parameters, like the distributions for household age and income, I chose to use an improved data source using similar methodology. These parameters in particular were calculated using the Family Resources Survey [39], which is an accredited official statistic in the UK in 2024, unlike the original latest Wealth and Assets Survey which was used to calculate the distributions for the 2011 target-year model [10, 40].

After we calculate this new parameter, I choose to validate each individual parameter in a three-step process. Firstly, I validate the parameter via comparing to external statistics I can find. For example, does the average income from our income distribution actually match the mean income in the UK in 2024? Secondly, I chose to face validate all of these calculated parameters using the calibration section of my ABM website, which is introduced in Section 6.1. Finally, I use my newly introduced validation loss function from Chapter 3 to compute the per-validation metric and total loss for each updated model version. Importantly, I treated a large change in loss following a single parameter update as a diagnostic warning rather than conclusive evidence that the updated estimate was invalid. I felt this caution is necessary because ABMs often induce noisy, non-linear and rugged parameter-to-output response surfaces, so a local loss spike may reflect stochastic variation, interactions with other parameters, or movement across an uneven region of the response surface rather than a clearly erroneous parameter estimate [7]. I investigated such cases further, including by checking the data source, estimation method and interactions with other recalibrated parameters.

This iterative recalibration strategy was useful throughout the 2024 update because it made the recalibration process more transparent, comparable and easier to debug. By storing intermediate model versions alongside their per-metric and total validation losses, I could compare the effect of each group of parameter updates against earlier versions rather than only assessing the final recalibrated model. This made it possible to observe validation progress over time, identify which updates improved or worsened particular model outputs, and investigate unexpected loss regressions at an early stage. The website visualisation further supported this process by allowing the original and updated model input parameters and output metrics to be compared directly, reducing the need for creating a new script manual inspection after every recalibration step. Overall, this produced faster development cycles and made errors in data extraction, parameter estimation or model behaviour easier to detect. However, intermediate versions must be interpreted cautiously, since many contain a mixture of 2011 and 2024 parameter values, meaning they do not represent a fully coherent target-year calibration. This staged approach also required parameter groups to be prioritised manually, which I theorised may miss interactions between parameters that only become visible when several updates are applied together.

5.2.2 Example recalibration

The Family Resources Survey (FRS) is an annually released officially accredited household-micro-data UK dataset focussed on income and household circumstances. The Wealth and Assets Survey (WAS) is a biannual survey conducted in the UK to assist in understand households' capacities in 'assets, savings, debt and planning for retirement' [41, p. 4]. In the original housing market ABM targeted to 2011, the WAS was used to estimate the age and gross-income distributions, with micro-data from *Wave 3* of the WAS, where data was collected in 2010-2012 [42]. As of 2024, this data is over 12 years out of date. In the model, a household's age, income, and wealth affect which houses they can afford to rent or mortgage. This in turn affects which households buy or rent, which renters become first-time buyers and which quality of houses households bid for.

For this example recalibration and data-source improvement, I aim to update the calibration to be the 2024 iteration of the Family Resources Survey (FRS) [39]. I chose to use the FRS as the data source for these distributions instead of the WAS as the FRS is a UK-wide dataset whereas the WAS only surveys Great Britain, the FRS is an accredited official dataset, whereas the WAS is not in its most recent release and finally the FRS has data from the 2024 target year but the WAS doesn't. The Office of National Statistics stated 'Data collected on the Wealth and Assets Survey covering the period April 2022 to March 2024 (Round 9) are currently undergoing post-collection data processing' and through correspondence with the ONS, I found they are planning to release this survey by the end of the year.

Household Age Calibration

Age is a particularly crucial factor in modelling the UK housing market, as it dictates a variety of factors in the model. At different ages, people earn different incomes and have varied wealth. This means different age groups are more or less likely to rent, own and occupy or buy to let, significantly influencing our model. The House of Commons states the UK population is currently ageing [43] and our model should ideally show shifts in these different age groups.

Within the model itself, an age distribution is used to maintain a constant overall age distribution by varying birth and death rates. This means for the duration of the simulation, the distribution of ages in the model stay consistent with the age distribution of the target year.

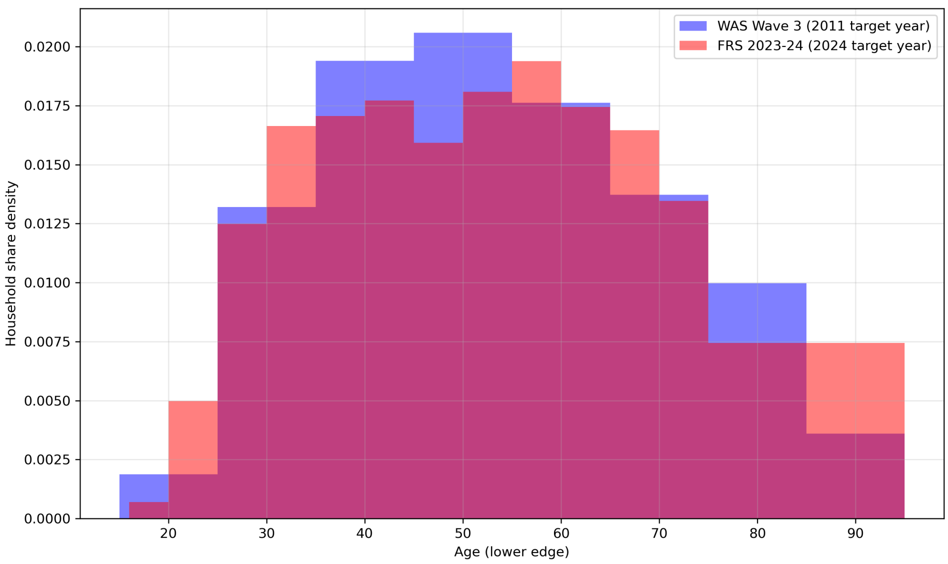

Figure 5.2: Age Distribution Bins between 2010-2012 and 2023-2024

Dataset	Period	Mean	Std. Dev.	Skew
WAS Wave 3	2011 target year	53.797	17.165	0.218
FRS 2023-24	2024 target year	54.649	18.481	0.223
Percent diff. (2024 vs 2011)	-	1.58%	7.67%	1.95%

Table 5.4: Age distribution data-source comparison summary statistics

Through updating our dataset from the WAS in 2010-2012 (Wave 3) to the FRS in 2024, as presented in Table 5.4 and Figure 5.2, average household age in the UK has increased by 1.58% in 10 years, from 53.797 to 54.649 years old, which is consistent with the UK population ageing. Interestingly, standard deviation of household age has increased significantly (7.67%). Perhaps this is due to the distribution more accurately reflecting the stronger first-time-buyer participation since 2011 [22]. Another likely reason for this is due to the granularity of the data sources used. The WAS Wave 3 dataset has nine age bins, whereas the FRS 2023-24 release has fifteen bins. Updating the model using this newer FRS data will lead to more households with an older age, which in turn will affect average wealth and income per household, which is explained further in section 5.2.2.

Gross-income by Age Calibration

In the model, households are assigned a fixed income percentile for life at their birth. Every month, the household is then assigned an income based on the household's age. Income in real-life, alongside wealth, determines the capacity of a household to afford mortgages or rent houses, the probability they become buy-to-let investors and the household wealth. I expect updating the WAS to a more recent dataset to increase average income at every age group due to nominal growth and steady inflation in the UK over time.

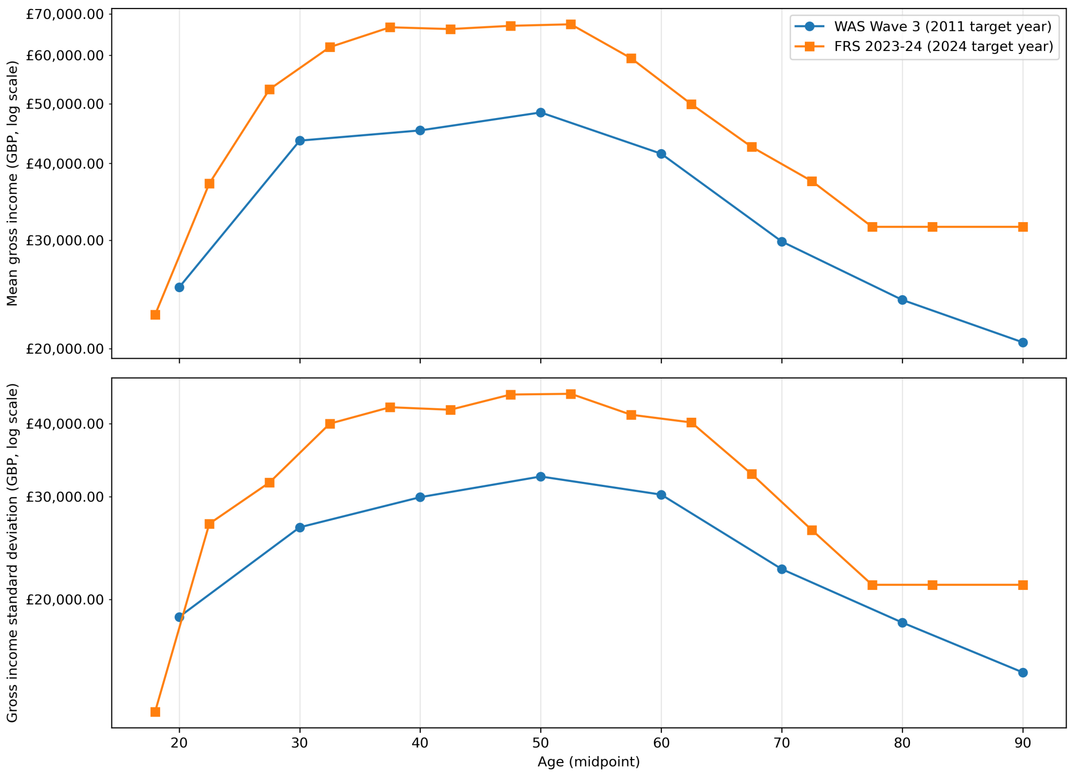

Figure 5.3: Joint gross-income by age distribution comparison: 2010-2012 vs 2023-2024

Dataset	Period	Mean	Std. Dev.	Skew
WAS Wave 3	2011 target year	£39,517.09	£29,021.00	1.649
FRS 2023-24	2024 target year	£53,220.88	£39,484.98	1.653
Percent diff. (2024 vs 2011)	-	34.68%	36.06%	0.25%

Table 5.5: Joint gross-income by age distribution summary statistics

Through updating our dataset from the WAS in 2010-2012 (Wave 3) to the FRS in 2024, presented in Table 5.5 and Figure 5.3, we find there is higher gross-income on average overall between datasets, with mean household gross income increasing by 34.68% in 10 years, from £39.5k to £53.2k, with a near identical 36.1% increase in standard deviation.

Updating the model using this dataset caused the model to reflect these changes in income distribution, which is reflected in Figure 5.4, where we compare the 2024 model income output to both the Wealth and Assets Survey round 8 validation dataset [44] (defined in Table B.1). We find model income outputs have shifted with a increase in distributional mean of £14.0k, or 35.7%. I choose to validate model outputs compared to the latest release of the Wealth and Assets Survey rather than the Family Resources Survey because the FRS is already used to recalibrate the income distribution. Using the WAS as a separate validation source provides a more independent test of whether the recalibrated model generalises beyond the calibration dataset. This is especially appropriate because the WAS is also the original data source used by Carro et al. for the 2011-calibrated model [10], which makes it useful for maintaining methodological continuity when comparing the original and updated model versions. However, I think this comparison should be interpreted with caution because, as previously mentioned, the latest available WAS data covers 2020-2022 rather than the 2024 target year. Some difference between the model output and the validation distribution are therefore expected, which is reflected in the 2024 output having a 15.4% higher mean than the validation dataset in Figure 5.4. We discuss the future work of updating the Wealth and Assets Survey to the not yet released Round 9 for this recalibration in Section 7.2.

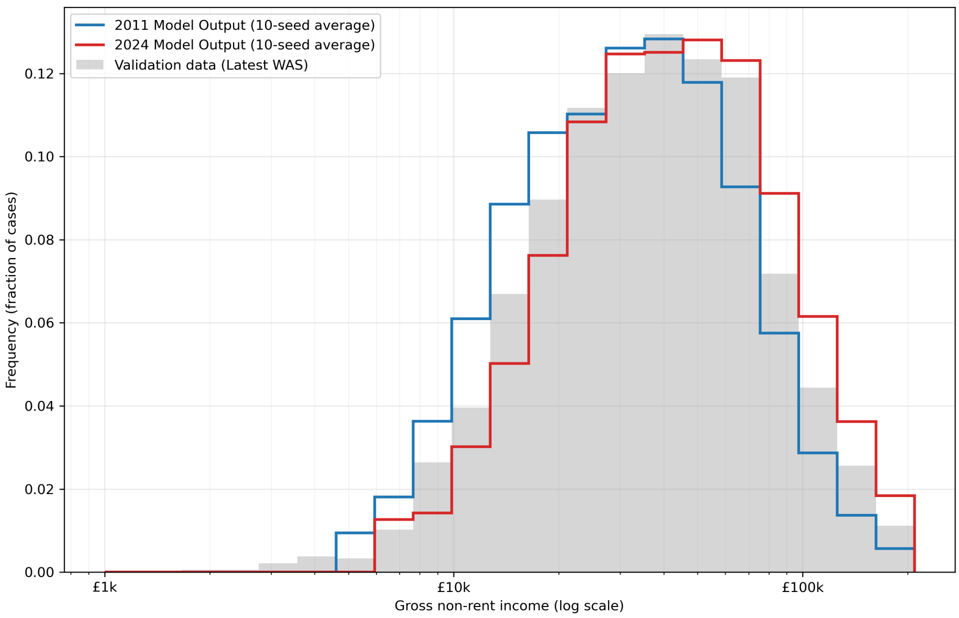

Figure 5.4: Household income distribution of 2024 model output versus validation data

5.2.3 Results

The example recalibrations in Subsection 5.2.2 are the details of two of complex distributional input parameters that needed to be empirically re-estimated and also had their dataset changes from the original parameters in Carro et al.'s paper [10]. In total, I recalibrated 59 out of the 75 total parameters to the state of the UK housing market in 2024. The 16 parameters that were not recalibrated in this effort were due to a lack of non-commercial or non publicly available data sources existing. Six parameters were not recalibrated due to lack of non-commercially available listing-level sale market data. Four parameters were not recalibrated due to lack of non-commercially available listing-level rental market data. Six parameters were not recalibrated due to the Bank of England's mortgage Product Sales Data being private confidential data. A full breakdown of the recalibration of every parameter is presented in Appendix Section C.2.

Via comparing the 2024 model to the state of the UK housing market in 2024 using the validation metric defined in Chapter 3, we yield validation loss results listed in Table C.4. We find the 2024 recalibrated model has a 3.75% higher composite validation loss than the 2011 target-year model. The 2024 recalibration significantly outperforms the loss on certain metrics such as a 100% reduction in interest rate spread loss, a 94.8% reduction in mean household debt-to-income loss and a significant 45.9% reduction in house cycle amplitude loss (HPI Std).

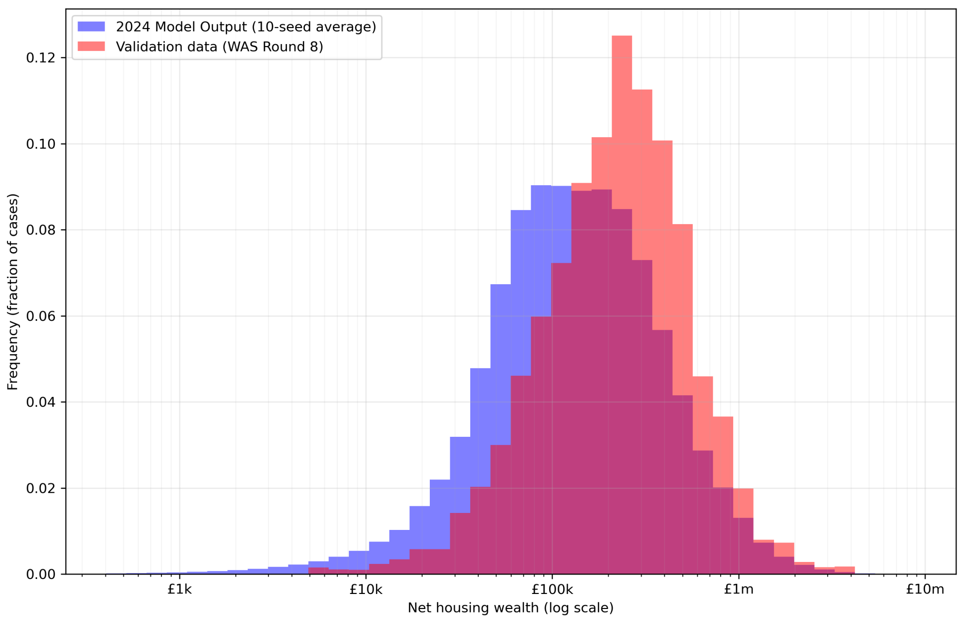

Figure 5.5: Household property wealth distribution of 2024 model output versus validation data

Although the 2024 recalibrated model performs well in some areas, it significantly underperforms in others. Four metrics, including HPI cycle period, income distributional realism, mortgage approvals and distributional housing wealth realism each have over and 100% increase in validation loss compared to the 2011 calibrated model. In particular, housing wealth has a 740% increase in loss compared to the 2011 model. Analysing housing wealth compared to validation data using Figure 5.5, we find the model's housing wealth output is far below the actual real-world distribution, with the 10 seed model mean being £96.2k lower than validation and standard deviation £73.0k lower.

Via data analysis on 10-seed 2011 and 2024 model outputs, we find that currently, although the 2024 model has mean house prices as £375k compared to the 2011 model's £225k, average house price sales are only £176k compared to the 2011 model's £170k. Moreover the 2024 has too much average owner-occupier debt-to-income with 117.3% compared to the 78.8% real-world 2024 statistic from the MLAR/ONS [45, 46]. I hypothesise that without sufficiently calibrated sale-market price-dynamics, due to lack of non-commercially available listing-level sale market data, this model cannot sufficiently transmit the recalibrated 2024 reference house-price distribution into realised transaction prices and household net housing wealth. In particular, unchanged legacy listing mark-up, price-reduction, bid-up, days-under-offer, and market-average-price-decay parameters may keep the effective sale-price process anchored close to the 2011 regime. Since model housing wealth is measured as expected sale value net of outstanding mortgage principal, the 2024 model is then doubly penalised: realised sale prices remain too low, while mortgage leverage and owner-occupier debt remain too high. This suggests that simply recalibrating the exogenous house-price distribution is insufficient unless sale-market price dynamics and credit constraints are recalibrated jointly.

Although I put in significant effort to recalibrate the majority of parameters in the model, I believe the model is not yet sufficiently recalibrated to 2024. In Section 7.2 I will describe the effort required to finish recalibrating the model sufficiently to 2024 and the post-Covid state of the UK housing market.

Chapter 6

Improving Accessibility of the ABM through a Graphical User Interface

As briefly described in Section 2.3, Carro et al.'s ABM is quite inaccessible for policymakers out-of-the-box for several reasons:

- Model runs are difficult to configure. User set parameters and central bank configurations must be configured manually via editing a text file called `config.properties`. Additionally the ABM must be run via command line, which is slow and tedious.
- Model outputs are recorded as raw semicolon-delimited CSV-style files. The main output file records aggregate model statistics at every time-step, whilst others are either sparse per-household snapshots, transaction-level event logs or multi-run core-indicator series without explicit time-step labels. This makes results difficult to interpret without additional post-processing code.
- Calibrated parameters are stored as raw values and often are hard to understand and quantify for users. For example, the distribution for house prices is stored as log scale (μ) and log shape (σ). Many of these parameters are therefore hard to interpret without additional assistance and graphical representation.
- Users must have a compatible Java runtime installed before the model can be built or run, adding another technical prerequisite before any experiments can be performed.
- Multi-run analysis is awkward in the ABM. Repeated simulations produce separate run-specific files, while some core indicators are written as separate per-indicator files with simulation runs arranged by row. This makes comparing runs, computing uncertainty bands, and summarising scenario results difficult without additional aggregation scripts.

Problem statement: This ABM remains inaccessible to non-technical policy users because configuration, execution, output interpretation, parameter understanding and multi-run analysis all rely on manual, code-based workflows.

To improve this inaccessibility, I decided to engineer a graphical user interface (GUI) to facilitate ABM experimentation and analysis. This interface was released both as an Electron-based Windows executable for local use and as a web application hosted on AWS [47, 38]. Currently this website has three sections which is the homepage shown in Figure 6.1, the calibration section of the website which is discussed in Section 6.1 and the experiment run part of the website which is discussed in Section 6.2. This dashboard was presented to engineers and analysts at the Bank of England during two in person meetings in March and May of 2026.

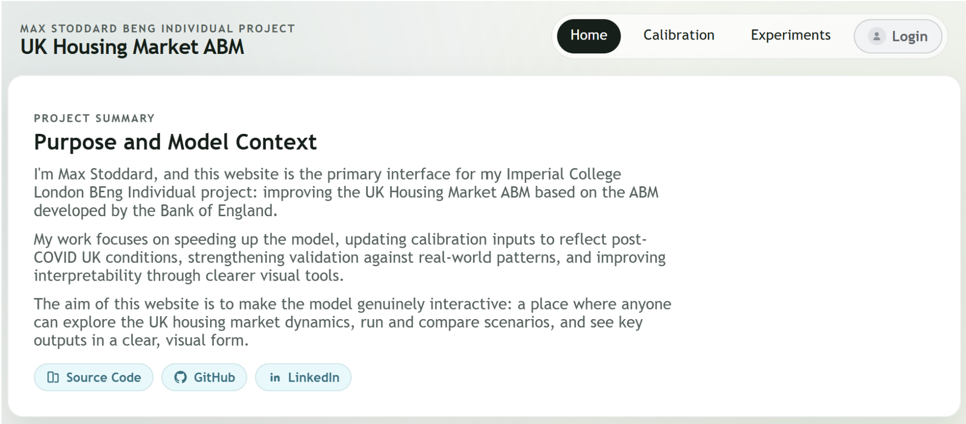

Figure 6.1: Homepage of the publicly hosted ABM website [38].

6.1 Website calibration section

The first part of the website I chose to create was the section to visualise the calibrated parameters. As mentioned in the start of Chapter 6, calibrated and estimated parameters are currently hard to visualise in the model without running external scripts due to storage as raw values in log-space. This makes analysis and face validation of new parameters difficult. All 75 parameters model parameters, detailed in Section C.2, are now visualisable via the ABM website/executable [47, 38].

In Figures D.1 and D.2, we give two example visualisations of parameter sets. The key features of this section of the website are:

- Raw parameters are combined into logical small groups of parameters. For example the house price distribution values presented in Figure D.1 are stored in the model's `config.properties` file as two separate parameters called `HOUSE_PRICE_SCALE`/`_SHAPE` which define the mean and std. dev. of this house price log-normal distribution. I instead present these as a singular house price distribution in actual monetary scale.

- I also allow comparison of parameters across different model 'versions'. We can directly compare the parameters of Carro et al.'s original ABM [10], the TuRBO optimised version of the ABM presented in Section 5.1 and the 2024 model recalibration presented in Section 5.2. Parametric distributional parameters, like the log-normal house price distribution in Figure D.1 are comparative via plotting both parameters sets on the same axis with median lines. Binned empirical distributions like the household wealth by gross income distributions, are presented as a heat-map for each version and a delta heat-map as shown in Figure D.2.

- All graphs are interactive via hovering which allows the user to not just visualise but interact with and see how different parameters are distributed over their domain. For example, as shown in Figure D.2, we can interact directly with the heat-maps via hovering over their grid to see probabilities and deltas.

- I also chose to expose the exact sources, source file locations in the repo, key in `config.properties`, as well as the exact methodology used to recalibrate each parameter when recalibrating from 2011 to 2024. This transparency and accessibility of methodology and recalibration logic because it makes the recalibrated model easier to inspect, reproduce and challenge which makes the model more transparent as a research artefact.

The calibration website section contributes to solving our problem statement by converting code-level model parameters into interpretable visual artefacts. Rather than requiring users to inspect raw values in the model's `config.properties` file or run external scripts, the website groups related parameters, transforms them into meaningful real-world scales and then presents them through interactive graphs. By enabling direct comparison between the model versions from Chapter 5 users can better understand the calibration process. Finally, by exposing ABM data sources and methodology per-parameter, the website makes the model easier to inspect and challenge.

6.2 Website experiments section

I then chose to create a section of the website focussed on running experiments. My idea is that running experiment and viewing results in a UI eliminates the need for users to have a compatible Java runtime installed to run the model and clearly aggregates and plots results without additional python scripting.

Across either experiment type, the experiment section of the website has the following features:

- Users can run experiments on Carro et al.'s original ABM model [10], the TuRBO optimised version of the UK ABM from Section 5.1 or the in progress 2024 recalibration of the UK ABM from Section 5.2.
- Users can vary a 'base policy set' from either 2011 or 2024, corresponding to macroprudential Bank of England policy in either of these year.
- Users can choose their desired simulation duration, seeds per sampled point, month rolling window period for core indicators, transaction recording starting time and which model output metrics are actually output and therefore visualised. Runs with multiple seeds have results averaged over runs.
- Experiments are runs on top of a cache optimised ABM ran using the parallelisation framework presented in Chapter 4. Users can select how many parallel workers they want to run the model with to align with their system.
- Results are visualisable in graphical and aggregate forms, as presented in Appendix Figures D.5 and D.6 for effective analysis, across mean, coefficient of variation and range.
- All run settings have information tooltips which when hovered over give more clarifying information about setting purpose.
- Experiments jobs can be queued, raw results can be downloaded or deleted and we can choose which results we choose to view. Experiment job execution statistics given such as run throughput, eta, elapsed time and progress via a progress bar.

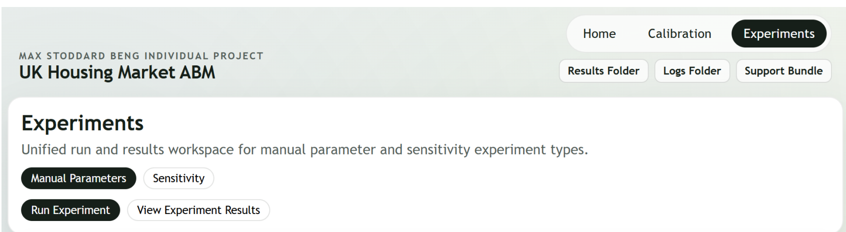

Figure 6.2: Selection of experiment types on ABM of type the household wealth distribution on the calibration section of the publicly hosted ABM website [38].

As shown in the selection part of the ABM website in Figure 6.2, I chose to divide the experimental part of the webpage into two experiment types which I call manual parameter experiments and sensitivity experiments.

Manual parameter experiments are experiments which generate runs with a single set of policy parameters across multiple seeds. For example a policy-maker may choose to test how tightening the 15% quota on the owner-occupier LTI flow limit cap affects model results [48], averaged over 20 seeds. Figure D.3 shows the run page settings for this experiment type. We can then directly view the results for one of these experiments or compare the results to another experiment. For example, a policymaker may choose to directly view the model outputs from the LTI flow limit tightening experiment, or compare to the a run with the current 15% quota policy. We allow the modification of any of the central bank policy parameters to whichever values the user selects. Comparison is made on an aggregate percentage difference and by plotting curves on the same graph. An example results page is shown in Figure D.5 comparing 20-seed runs of the original 2011 model to the optimised model from Section 5.1.

Sensitivity experiments are ran on a range of values for one parameter or policy. For example, a policymaker may choose to explore how varying the LTI flow limit cap across a variety of values from 0% to 30% affects model outputs. Figure D.4 shows the run page settings for this experiment type. The website then allows analysis into which model outputs are most sensitive to changes in these parameters and how each of these parameters actually varies most across seeds. Sensitivity experiments are not comparable between model versions as we aim to analyse the effect of modifying one parameter across a variety of values. Figure D.6 show an example results page where we choose to vary the LTI flow limit cap for both FTBs and HMs from 0% to 30%. A tornado plot and per-indicator delta trend is plotted as well as the full results table.

6.3 Windows executable version

To make this website as accessible as possible, I chose to first package the website as a Windows executable. To do this, I first build the frontend React dashboard using Vite and the Express API server and then package the actual ABM written in Java into a Java executable `.jar` file, alongside all dependencies and a bundled Java virtual machine. All of these files are then bundled using electron targeted at `x86`. I chose to automate this process alongside smoke checks into GitHub's continuous integration and deployment (CI/CD) platform called *GitHub Actions*. I then release these executables via GitHub releases at `https://github.com/max-stoddard/uk-housing-market-abm/releases`. I chose to store model run results and logs under Electron `userData`, in order to preserve this data across updates and uninstalls. In this version, the ABM is run directly on device.

Bundling the project in this way allows anyone with a default configured windows PC to easily access, run and experiment with this ABM, without any external install requirements, instead of needing advanced knowledge of command line and installing a multitude of dependencies.

6.4 Cloud website version

Additionally to packaging this website as an executable, I chose to also host this website remotely using Amazon Web Services. The architecture I chose for this cloud application is presented in Figure 6.3. I store the website assets in a private Simple Storage Service (S3) bucket, then CloudFront handles user HTTP requests globally using these private assets. Amazon's Application Load Balancer (ALB) gives the API a stable endpoint and routes requests to any Elastic Container Service (ECS) Fargate tasks alive, of which currently only one exists. ECS Fargate acts as the backend API handling stateful dashboard requests like experimental runs. A lightweight Elastic Compute Cloud (EC2) runs actual experiments and produces results into a separate S3 bucket for handling experiment run data.

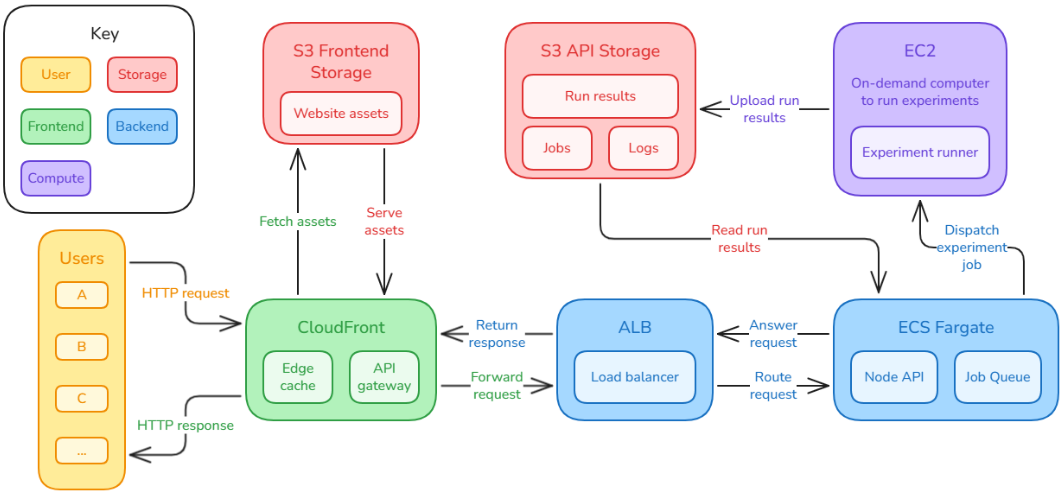

Figure 6.3: ABM Website Cloud Architecture Diagram

This architecture was designed to support future scalability. Although the current deployment uses a single API service and one EC2-based experiment runner, the experiment execution layer could be scaled by adding more, larger EC2 instances. Ideally, each of these instances would provide at least 32 vCPUs, allowing the application to better exploit the throughput gains from the parallelisation framework introduced in Chapter 4 and to support multiple concurrent experiment runs via scaling across multiple machines. If API request handling later becomes a bottleneck, the API layer could be scaled independently by increasing the number of ECS Fargate tasks behind the ALB. The static frontend layer, served through CloudFront and S3 already scales up globally and the ALB would naturally distribute API traffic across the available ECS tasks.

Additionally, since this website is hosted publicly, I employ multiple cloud security features. Users must access the site through CloudFront via HTTPS, the frontend S3 bucket is private and only accessible by CloudFront; no other computers and finally all experiment running and write actions on the website require login credentials.

Overall, the cloud deployment helps solve the problem statement by making the ABM accessible through a secure browser-based interface, removing the need for local Java setup, command-line execution or manual output handling, while supporting future multi-user scaling.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

Agent-based models of housing markets are valuable tools because they can represent the diversity of different household groups, such as renters, landlords or owner occupiers. These strengths are particularly important in the UK housing market, where changes in credit conditions, affordability and housing wealth affect households, lenders and policymakers simultaneously. However, full-scale ABMs are only practically useful if they can be validated systematically, recalibrated when market conditions change, repeatedly runnable at a feasible cost and inspected by users who are not necessarily familiar with the model's implementation. Carro et al.'s existing UK housing market ABM [10] provides a strong modelling foundation, but its practical use is limited by the cost of repeated simulation, the difficulty of comparing model versions, its 2011 calibration target and its code-based execution and analysis workflow. In this project I chose to address these limitations through a combined methodological, computational, empirical and engineering contribution. Rather than treating validation, performance, calibration and accessibility as separate issues, I aimed to improve the wider workflow by which the model can be assessed, optimised, updated and used. My central aim was to make future model development more measurable, reproducible and practically usable.

Firstly, I introduced a target-metric validation loss that compares model output against twenty UK housing- and mortgage-market indicators factoring in central accuracy, reliability and between-seed dispersion. This provided a common dimensionless framework for comparing model versions at a version level and at the level of these individual indicators. Secondly, I improved computational performance through a landlord rental-income cache optimisation and a parallel batch-running framework, achieving a 6.13-fold increase in model run throughput on realistic experimental workloads, corresponding to a 83.7% runtime reduction. Thirdly, I used the validation loss as the objective for Trust Region Bayesian Optimisation, improving the 2011 target-year model's validation loss by 7.79% whilst requiring 99.3% fewer model runs than the original grid-based calibration. Fourthly, I recalibrated all 59 of the model's 75 parameters for which suitable public data were available, producing a partial 2024 target-year version of the model. Finally, I improved accessibility by engineering a graphical user interface released both as a Windows executable and as a cloud-hosted web application, allowing users to visualise parameters, compare model versions, run experiments and analyse results without command-line execution or bespoke post-processing scripts. Taken together, these contributions make the ABM more useful as a research and policy-analysis tool. The validation metric makes model changes easier to evaluate; the performance improvements make repeated validation, calibration and sensitivity analysis more feasible; the 2024 recalibration shows how the model can be updated to reflect a more recent housing-market state; and the interface makes the model easier to inspect, challenge and use. The project therefore strengthens not just the model itself, but the surrounding process required to develop, compare and apply it in future work.

However, this project does not make the ABM a complete representation of the contemporary UK housing market. The 2024 recalibration remains incomplete because several parameters still require confidential or commercially restricted data. The model also retains structural limitations, including simplified banking behaviour, the lack of default and foreclosure dynamics and continued downsampling of the household population. Future work should therefore focus on completing the 2024 baseline, extending the realism of the model and improving computational scalability.

7.2 Future work

The most immediate area for future work is completing and strengthening the 2024 recalibration. For this project I updated all parameters for which suitable public data sources were available, but sixteen parameters could not be recalibrated due to unavailable non-commercial sale-listing or rental-listing data or the FCA's private loan-level mortgage Product Sales Data. Future work should therefore incorporate these missing data sources whenever access becomes possible, in order to calibrate sale-market and rental-market price dynamics, and mortgage-product parameters. This is particularly important because I found the 2024 model suggests that updated household-level inputs alone are not sufficient to fully transmit the contemporary house-price distribution into realised transaction prices and household housing wealth. The recalibration should also aim to integrate and use the Wealth and Assets Survey Round 9, scheduled to be released by the end of 2027[1] and was collected from April 2022 to March 2024, since the current validation uses the latest available release rather than a survey exactly matching the 2024 target year [41].

A second direction future work could take is improving the structural realism of the model. The current model retains several simplifying assumptions from the original UK housing ABM, including the absence of explicit mortgage default, arrears, foreclosure and lender losses. Adding these mechanisms would make the model more suitable for analysing financial-stability risks, especially under adverse affordability or interest-rate scenarios. The model also uses a simplified banking sector, where mortgage lending is represented through an aggregate lender rather than heterogeneous banks with different balance sheets, risk-tolerance and lending constraints. Introducing multiple heterogeneous banks would allow future experiments to analyse how macroprudential policies affect not only households, but also different types of lenders and their exposure to risky mortgage lending [9, 10].

A third direction future work could take is further improving computational scalability and performance. The caching and parallelisation work in this project significantly reduced batch runtime, but further gains are likely possible through deeper profiling, additional algorithmic optimisations and more scalable execution infrastructure. Future work could examine whether further bottlenecks emerge at larger population sizes, whether additional repeated calculations can be cached safely, and whether cloud worker pools can be scaled dynamically for large validation or calibration jobs. This would be particularly useful for future recalibration work, where repeated model evaluations are expensive.

[1] This date was obtained through direct correspondence with `wealth.and.assets.survey@ons.gov.uk`

Declarations

Use of generative AI

I acknowledge the use of OpenAI's ChatGPT and Codex (OpenAI, https://chatgpt.com and https://chatgpt.com/codex), both utilising the GPT-5.5 and GPT-5.4 models at `xhigh` reasoning (https://developers.openai.com/api/docs/models) during this project. These tools were used as research and development companions throughout the project to support reasoning, develop understanding, identify potential flaws in my methods, assist with LaTeX syntax and assist with implementation. This use of GenAI for this project was discussed with and confirmed as permitted by my supervisor, Ce Guo. I manually reviewed all AI-assisted suggestions before use. Code suggestions were checked against intended behaviour, tested where relevant, and rejected or adapted where they conflicted with my design or source evidence.

No AI-generated text has been included in this report, which was written entirely by me, however, ChatGPT (OpenAI, https://chatgpt.com) utilising the GPT-5.5 model at `xhigh` reasoning was used to identify spelling and grammatical errors. Diagrams 2.1 and 6.3 were made manually using https://excalidraw.com/.

Finally, I confirm that the final submitted work, including all interpretations, decisions, and conclusions, is my own.

Ethical considerations

All data used to validate or calibrate this model has been confirmed to be usable in academic context. I ensured no raw datasets are published in my GitHub repo; data is only stored in aggregated model parameter form. Ethically, this project aims to benefit society at large. By improving this model, economists and researchers can make more informed, evidence-based, and robust policy assessments relating to the UK housing market, with potential benefits for housing affordability, rental market stability, access to home ownership, and the broader cost of living in the UK.

Sustainability

All experiments ran on this model were run on my local, 20-core, Intel-based laptop. Experimental sizes were selected primarily to ensure validity and robustness, while avoiding unnecessarily large runs that would increase computational cost and energy consumption without materially strengthening the conclusions. Efforts were made to improve the computational efficiency of running this model, such as implementing the model caching optimisation in Section 4.1 and utilising the modern TuRBO in Section 5.1 over less computationally efficient techniques. I deliberately designed my cloud-hosted ABM website, introduced in Section 6.4, to be minimal and resource-efficient, including only the components required to support deployment to reduce complexity and energy consumption.

Availability of data and materials

The source code for this project is publicly accessible in the GitHub repository at https://github.com/max-stoddard/uk-housing-market-abm. This repo is built from a clone of Carro et al.'s housing market ABM: https://github.com/INET-Complexity/housing-model [10]. In my repo, model source code is contained in `src/main/`, the project GUI is contained in `dashboard/`. Experiment and validation scripts can be found under `scripts/`. The model versions from Chapter 5 can be found under `input-data-versions/`.

Model calibration data sources were accessed via the UK Data Service under project ID 290608.

Page Count

Excluding my title page, table of contents, abstract, declarations and appendices, this report is 46 pages of content with 2.5cm margins and font size 11. This is below the 60 pages of content limit.

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Appendix A

Acronyms

Acronym	**Description**
BoE	Bank of England
BMS	BM Solutions (Lloyds Banking Group)
CML	Council of Mortgage Lenders
FCA	Financial Conduct Authority
FPC	Financial Policy Committee
FRS	Family Resources Survey
HMLR	HM Land Registry
MLAR	Mortgage Lenders and Administrators Return
ONS	Office for National Statistics
PSD	FCA Mortgage Product Sales Data
UKF	UK Finance
WAS	Wealth and Assets Survey

Table A.1: Definitions of data source acronyms

Acronym	**Description**
BTL	Buy-to-let
DTI	Debt-to-income ratio
DSTI	Debt Service-to-Income
FTB	First-time buyer
HH	Household
HM	Home mover
HPI	House Price Index
JSD	Jensen-Shannon divergence
NSA	Non-seasonally adjusted
OO	Owner-occupier
pp	Percentage points
PIR	Price-to-income ratio
RPI	Rental Price Index
SA	Seasonally adjusted
Std	Standard deviation

Table A.2: Definitions of other acronyms used in this appendix

Appendix B

Validation

B.1 Validation Target Metric Data Sources

Metric	2011 source(s)	2024 source(s)
Mortgage Approvals	BoE [49]	BoE [49]
Housing Transactions	BoE [49]	BoE [49]
Advances to FTB	CML[1]	UKF [50]
Advances to HM	CML[1]	UKF [50]
Advances to BTL	CML[2]	UKF[3]
HH DTI	BoE [49]	BoE [49]
House PIR	BoE [49]	BoE [49]
House Price Growth	BoE [49]	BoE [49]
HPI Mean	HMLR [51]	HMLR [51]
HPI Std	HMLR [51]	HMLR [51]
HPI Cycle Period	HMLR [51]	HMLR [51]
RPI Mean	ONS [52]	ONS [52]
HH Owning Share	FRS [53]	FRS [54]
HH Private Renting Share	FRS [53]	FRS [54]
OO DTI	MLAR [45], ONS [46]	MLAR [45], ONS [46]
Rental Yield	BMS[4]	UKF[3]
Interest Rate Spread	BoE [49]	BoE [49]
Income Realism[5]	WAS [42]	WAS [44]
Housing Wealth Realism[5]	WAS [42]	WAS [44]
Financial Wealth Realism[5]	WAS [42]	WAS [44]

[1] Verified through PropertyWire [55] and Mortgage Finance Gazette [56] secondary reports quoting Council of Mortgage Lenders first-time-buyer and home-mover purchase-loan totals.
[2] Verified through a Mortgage Strategy [57] secondary report quoting the Council of Mortgage Lenders buy-to-let house-purchase total.
[3] Calculated using UK Finance Buy-to-let Mortgage Market summary-panel values from four quarters of 2024 data [58, 59, 60, 61].
[4] Verified through contemporaneous PropertyWire reports quoting BM Solutions rental-yield figures: the target uses the reported 2011 UK average gross yield of 6.1% [62], with June 2011 [63] and Q4 2011 figures [64] used as cross-checks.
[5] For these household realism target metrics, we compare model outputs to target histograms computed from Wealth and Assets Survey data via Jensen-Shannon divergence.

Table B.1: Validation sources for both 2011 and 2024 target years

B.2 Validation Target Metric Values

Table B.2 describes the validation target data associated with all 20 validation target metrics in both the target years 2011 and 2024. **Metric** is the name of the specific validation target metric. See tables A.2 for acronym definitions if unclear. **2011 Value** and **2024 Value** are the source values for the target years of 2011 and 2024 respectively. Source values are calculated directly from the sources in table B.1. **Range** is the target range for that metric. Where possible we derive the within-yearly minimum-maximum range for a metric from a source series. If this is not possible, we define the target range to be ±5% of the target value as a conservative rounding of the average target-band half-width of the derived metrics of ±4.27%. This half-width calculation excludes signed additive family (**add**) metrics as percentage deviations near-zero are not substantively meaningful. **Family** is the metric families defined in section 3.1.2 and define how validation loss is calculated for that metric.

Metric	Unit	2011 Target	2024 Target	Range	Family
Mortgage Approvals	count/month	49,286.3	61,325	Derived[1]	**pos**
Housing Transactions	count/month	73,551.7	84,200	Derived[1]	**pos**
Advances to FTB	count/month	16,083.3	27,833.3	±5%	**pos**
Advances to Home Movers	count/month	26,375	24,000	±5%	**pos**
Advances to BTL	count/month	7,000	5,171.25	±5%	**pos**
HH DTI	%	162.938	133.8	Derived[1]	**pos**
House PIR	ratio	4.24	5.4	Derived[1]	**pos**
House Price Growth	%	-0.397	1.1	Derived[1]	**add**
HPI Mean	index[4]	0.992284	1.01969	±5%	**pos**
HPI Std	index[2]	0.27667	0.27667	±5%	**pos**
HPI Cycle Period	months[3]	171.333	167.5	±5%	**pos**
RPI Mean	index[5]	1.01096	1.04042	Derived[1]	**pos**
HH Owning Share	%	66	65	±0.5pp[6]	**share**
HH Private Renting Share	%	17	19	±0.5pp[6]	**share**
OO DTI	%	99.2017	78.7969	Derived[1]	**pos**
Rental Yield	%	6.1	6.9275	See note[7]	**pos**
Interest Rate Spread	pp	2.48008	0.541198	Derived[1]	**add**
Income Realism	-	JSD	JSD	≤ 0.12	**low**
Housing Wealth Realism	-	JSD	JSD	≤ 0.12	**low**
Financial Wealth Realism	-	JSD	JSD	≤ 0.12	**low**

1 Range is derived from the minimum-maximum range of the tracked source series to reflect within-year variation in the source data.
2 HPI standard deviation uses the shared HM Land Registry UK IndexSA long-run population standard deviation from January 2005 to December 2024, rebased to January 2005 = 1.0; it is not a 2011-only standard deviation. This is in order to give a long-run average amplitude metric.
3 HPI cycle period is derived from the HM Land Registry UK Index history through 2011-12 using a 12-month moving average, log de-trending, and a fast Fourier transform dominant-cycle search over 60 to 240 months.
4 HPI mean uses the HM Land Registry UK seasonally adjusted house price index rebased to 1.0 at the target year's January value: January 2011 for the 2011 target and January 2024 for the 2024 target.
5 RPI mean uses the ONS private rents Great Britain index rebased to 1.0 at the target year's January value: January 2011 for the 2011 target and January 2024 for the 2024 target.
6 FRS tenure tables publish whole-percent shares only, so this is a rounded-value interval.
7 The 2011 rental-yield target uses a fixed ±15% tolerance around the best-available annual BMS UK rental-yield metric. The 2024 target uses UK Finance quarterly is derived from the minimum-maximum range of the tracked source series to reflect within-year variation in the source data.

Table B.2: Validation metric values and acceptance ranges

B.3 Per-metric validation metric stability

Figure B.1 shows the validation loss, defined in Chapter 3, comparing the TuRBO output-calibrated 2011 target-year version of the UK housing ABM (defined in Section 5.1) to Carro et al.'s original UK Housing ABM [10]. Across all 20 target metrics, we test the stability of this comparative loss under different weights for target-band reliability and between-seed dispersion weights, r and d respectively, in the generic loss equation 3.2.

Blue values indicate that the metric improved between the optimised version compared to the original version, red indicate they regressed. We test stability by ensuring sign stability on a per-metric basis whilst varying these candidate loss weights r and d.

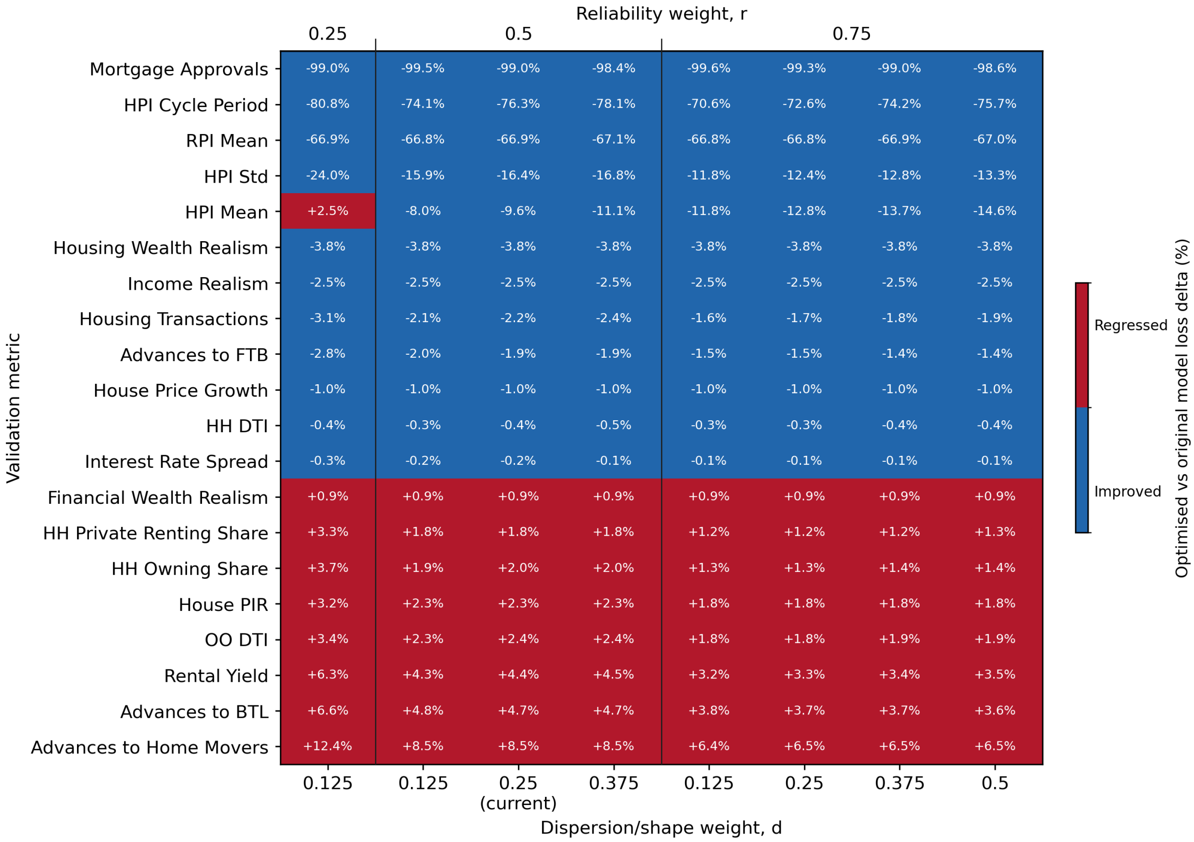

Figure B.1: Sign-stability of validation metric losses under alternative loss-weight settings.

Appendix C

Calibration

C.1 TuRBO Optimised 2011 Model Loss

Table C.1 summarises the validation loss of the new TuRBO optimised version of the housing market ABM compared to the original model validation defined in Section 3.3. 2011 target-year UK housing market ABM [10] introduced in Section 5.1. For each of the twenty-one target validation metrics defined in Table B.2, for the optimised ABM **Loss** indicates the validation loss for this particular metric, **Loss Delta** indicates the difference in loss between this optimised version and the original model and **Loss Delta %** is this difference as a percentage of the original calculated via computing $\frac{\mathcal{L}_{x,\text{optimised}} - \mathcal{L}_{x,\text{original}}}{\mathcal{L}_{x,\text{original}}}$ as a percentage.

Metric	**Loss**	**Loss Delta**	**Loss Delta %**
Mortgage Approvals	0.00270	-0.256	-99.0%
HPI Cycle Period	0.100	-0.321	-76.3%
RPI Mean	0.153	-0.309	-66.9%
HPI Std	0.517	-0.101	-16.4%
HPI Mean	0.514	-0.0547	-9.61%
Housing Wealth Realism	0.166	-0.00647	-3.75%
Income Realism	0.0843	-0.00214	-2.47%
Housing Transactions	0.749	-0.0172	-2.24%
Advances to FTB	0.798	-0.0156	-1.92%
House Price Growth	0.0113	-0.000114	-0.995%
HH DTI	0.981	-0.00393	-0.399%
Interest Rate Spread	0.641	-0.000972	-0.151%
Financial Wealth Realism	0.196	+0.00173	+0.890%
HH Private Renting Share	0.539	+0.00946	+1.79%
HH Owning Share	0.538	+0.0104	+1.97%
House PIR	0.930	+0.0209	+2.30%
OO DTI	0.840	+0.0196	+2.38%
Rental Yield	0.826	+0.0347	+4.38%
Advances to BTL	0.979	+0.0443	+4.75%
Advances to Home Movers	0.857	+0.0671	+8.50%
Total Composite Loss	0.521	-0.0441	-7.79%

Table C.1: Validation loss comparison between TuRBO optimised and original 2011 target-year ABMs

C.2 2024 Recalibrated Housing ABM Parameters

Parameter	2011 value	2024 value	2024 source(s)
Central Bank Policy			
Central Bank			
Hard maximum LTV ratio			
First-time buyers	0.95	0.95	N/A[1]
Home movers	0.9	0.95	N/A[1]
BTL investors	0.8	0.85	N/A[1]
Soft maximum LTI ratio			
First-time buyers	5.4	4.5	BoE [48]
Home movers	5.6	4.5	BoE [48]
Hard maximum home buyers DSTI ratio	0.4	0.9999	N/A[1]
Hard minimum ICR for BTL investors	1.2	0.0	N/A[1]
Exogenous policy rate	0.005	0.051083	BoE [75]
Population			
Number of UK dwellings	22,626,000	30,676,974	ONS [76, 77, 78, 79]
Number of UK households	26,442,100	28,609,000	ONS [80]
Household Characteristics			
Age	Est. dist.[2]	Est. dist.[2]	FRS [81]
Gross monthly income	Est. dist.[2]	Est. dist.[2]	FRS [81]
Financial wealth	Est. dist.[2]	Est. dist.[2]	WAS [40]
Consumption Behaviour			
Essential non-housing consumption fraction	0.66	0.651012	LCFS [82]
Maximum desired consumption fraction	0.17	0.196466	LCFS [82]
Household Expectations			
House price growth expectation			
Multiplier factor	0.44	0.28879	NMG, HMLR [83, 51]
Constant	-0.007	-0.005959	NMG, HMLR [83, 51]
Time period (years)	2	2	N/A[3]
Purchase Behaviour			
Desired purchase price			
Scale	42.9036	4.39575	PSD, HMLR [84, 51]
Exponent	0.78917	1	PSD, HMLR [84, 51]
Mean of normal noise	-0.017687	0	PSD, HMLR [84, 51]
Std. dev. of normal noise	0.410377	0.2	PSD, HMLR [84, 51]
Desired down-payment			
First-time buyers			
Scale	10.35	10.6566	PSD [84]
Shape	0.898	1.05251	PSD [84]
Home movers			
Scale	11.15	11.6263	PSD [84]
Shape	0.958	0.875107	PSD [84]
Buy-to-let			
Scale	11.15	11.6263	N/A[4,5]
Shape	0.958	0.875107	N/A[4,5]
Buy-to-let down-payment fraction			
Mean	0.34	0.34	N/A[4, 5]
Standard deviation	0.15	0.15	N/A[4, 5]
Sale Behaviour			
Holding period of owner-occupied houses	17.0	17.2	EHS [85]
Initial sale price mark-up	Est. dist.[2]	Est. dist.[2]	N/A[4]
Sale price reduction			

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Parameter	2011 value	2024 value	2024 source
Monthly probability	0.070262	0.070262	N/A[4]
Mean percentage reduction	1.45316	1.45316	N/A[4]
Std. dev. of percentage reduction	0.706952	0.706952	N/A[4]
Buy-to-let Behaviour			
Probability to receive BTL flag			
Raw probability	Est. dist.[2]	Est. dist.[2]	WAS [44]
Buy-to-let motivation			
Rental-income-driven probability	0.4927	0.401857	NMG [83]
Capital-gains-driven probability	0.1458	0.209303	NMG [83]
Mixed probability	0.3615	0.38884	NMG [83]
Rental Behaviour			
Desired rental price			
Scale	17.2166	18.1279	NMG[6] [83]
Exponent	0.346372	0.3371	NMG[6] [83]
Tenancy length			
Minimum	12	6	EHS [85]
Maximum	24	18	EHS [85]
Rent-out Behaviour			
Initial rent price mark-up	Est. dist.[2]	Est. dist.[2]	N/A[4]
Rent price reduction			
Monthly probability	0.105704	0.105704	N/A[4]
Mean percentage reduction	1.65587	1.65587	N/A[4]
Std. dev. of percentage reduction	0.785542	0.785542	N/A[4]
Housing Market			
Distribution of house prices			
Scale	12.1186	12.5352	HMLR[6] [86]
Exponent	0.641448	0.77434	HMLR[6] [86]
Distribution of rental prices			
Scale	6.26469	6.48827	NMG[6] [83]
Exponent	0.635275	0.803183	NMG[6] [83]
Rent gross yield	0.05	0.058134	ONS, HMLR [52, 51]
Bid-up parameter	1.07464	1.07464	N/A[4]
Days under offer	3.0	3.0	N/A[4]
Bank Behaviour			
Hard maximum LTV ratio			
First-time buyers	0.9	0.95	N/A[7]
Home movers	0.9	0.95	N/A[7]
BTL investors	0.75	0.85	N/A[7]
Hard maximum LTI ratio			
First-time buyers	5.4	5.4	N/A[7]
Home movers	5.6	5.6	N/A[7]
Hard maximum home buyers DSTI ratio	0.4	0.4	N/A[7]
Hard minimum ICR for BTL investors	1.2	1.25	Postulated[8]
Mortgage duration	25	32	PSD [84]
Retirement age	65	75	Postulated[9]
Elasticity of interest rate to credit	1.33×10^{-5}	5.47199×10^{-7}	BoE [49, 87]
Bank initial interest rate	0.035	0.056495	BoE [49]
Bank initial credit supply	244	704.939	BoE, ONS [87, 80]
Government Policy			
Income tax	Banded[10]	Banded[10]	GOV.UK [88]
National Insurance contributions	Banded[10]	Banded[10]	GOV.UK [89]
Personal allowance	£7,475.00	£12,570.00	GOV.UK [90]
Government income support	£445.80	£616.42	GOV.UK [91]

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Parameter	2011 value	2024 value	2024 source
Output-calibrated Parameters			
Psychological cost of renting	0.4	0.35	Calibrated[11]
Sensitivity parameter	0.001	0.0011	Calibrated[11]
Probability to receive BTL flag			
Probability adjustment	1.76	1.75	Calibrated[11]
Sensitivity of buy and sell decisions	100	110	Calibrated[11]
Weight on market vs segment house prices	0.5	0.5	Calibrated[11]

[1] This central bank policy parameter is set to a maximum/minimum limit or its corresponding bank behaviour parameter as to be non-binding, as the BoE does not currently impose this macroprudential policy.
[2] This parameter is an estimated distribution rather than a numeric value.
[3] This parameter was not modified as the postulation from the 2011 target-year model still holds [10].
[4] This parameter was not recalibrated to 2024 due to lack of open-source listing-level sale market or rental market data.
[5] BTL down-payment fraction parameters (mean and std. dev.) have been partially migrated to a HPI-scaled log-normal draw, equivalent to the FTB and HM desired down-payment shape and scale. This can be activated in by setting the `enableBTLDownpaymentLognormal` flag in the repo's `config.properties` file to true. By default this migration is turned off as there is insufficient data to calibrate the new BTL scale and shape parameters[4], hence we set these parameters equal to HM desired down-payment parameters as placeholders until calibration is done.
[6] Calibration fidelity was improved by fitting these parameters using survey weights and a log-space power-law fit to handle upper-bound risk and high-rent bias, rather than unweighted level-space.
[7] This parameter was not recalibrated to 2024 due to inaccessibility to access the private micro-data of the Bank of England's mortgage Product Sales Data.
[8] All 5 top BTL mortgage providers (Nationwide (TMW), Lloyds (BMS), NatWest, OneSavings Bank, Coventry) have 125% as their minimum standard lower-rate tax ICR [65, 66, 67, 68, 69].
[9] Bank retirement age is used as an owner-occupier repay-by constraint and as a BTL new-mortgage approval cut-off. For the top 5 mortgage providers in the UK, 75 is the modal threshold and appears as five of eight thresholds across application/repay-by age caps. [70, 71, 72, 73, 74].
[10] This parameter has income based bands rather than a numeric value.
[11] This parameter is calibrated using TuRBO output calibration, explained further in Section 5.1.

Table C.3: Recalibrated parameter values for the 2024 recalibrated version of the UK housing market ABM

C.3 2024 Recalibrated Housing ABM Validation

Metric	**2011 Loss**	**2024 Loss**	**Loss Delta**	**Loss Delta %**
Interest Rate Spread	0.642	0.0000917	-0.642	-100%
HH DTI	0.985	0.0516	-0.934	-94.8%
House PIR	0.909	0.363	-0.546	-60.1%
HPI Std	0.618	0.334	-0.284	-45.9%
RPI Mean	0.462	0.260	-0.202	-43.8%
House Price Growth	0.0114	0.00680	-0.00464	-40.5%
Advances to FTB	0.814	0.537	-0.277	-34.0%
Advances to Home Movers	0.790	0.632	-0.157	-19.9%
Financial Wealth Realism	0.194	0.167	-0.0273	-14.1%
HPI Mean	0.569	0.560	-0.00877	-1.54%
HH Private Renting Share	0.530	0.523	-0.00713	-1.35%
Rental Yield	0.792	0.831	+0.0398	+5.02%
OO DTI	0.821	0.881	+0.0607	+7.39%
Housing Transactions	0.767	0.887	+0.120	+15.6%
HH Owning Share	0.528	0.618	+0.0906	+17.2%
Advances to BTL	0.934	1.90	+0.964	+103%
HPI Cycle Period	0.421	0.866	+0.445	+106%
Income Realism	0.0865	0.191	+0.105	+121%
Mortgage Approvals	0.259	0.672	+0.413	+160%
Housing Wealth Realism	0.172	1.45	+1.28	+740%
Total Composite Loss	0.565	0.586	+0.0212	+3.75%

Table C.4: Validation loss comparison between original 2011 target-year ABM and 2024 recalibrated target-year ABM

Appendix D

Graphical User Interface

Figure D.1: Graphical visualisation of house price distribution parameters on the calibration section of the publicly hosted ABM website [38].

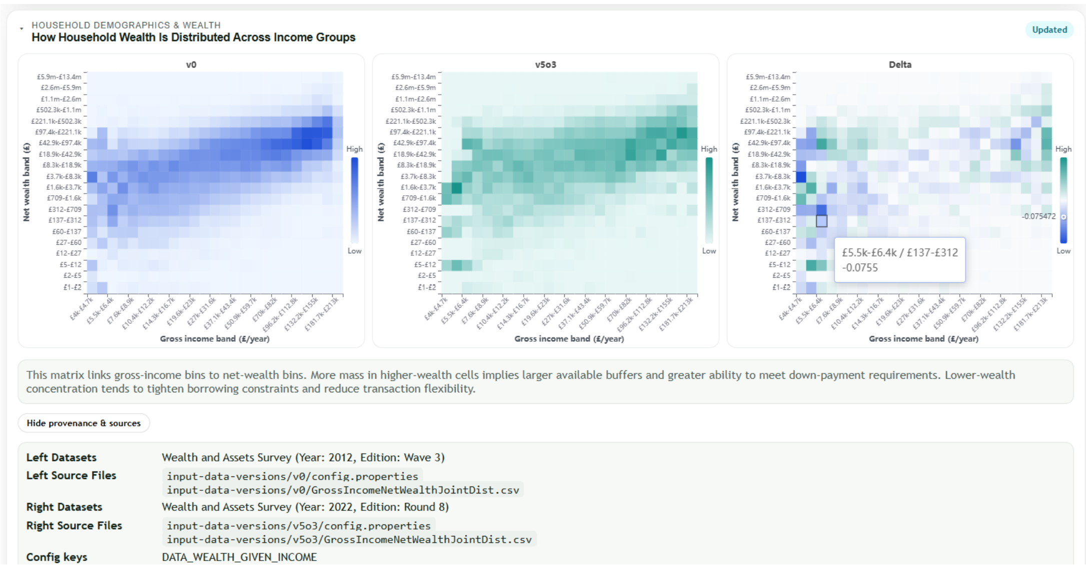

Figure D.2: Graphical visualisation of the household wealth distribution on the calibration section of the publicly hosted ABM website [38].

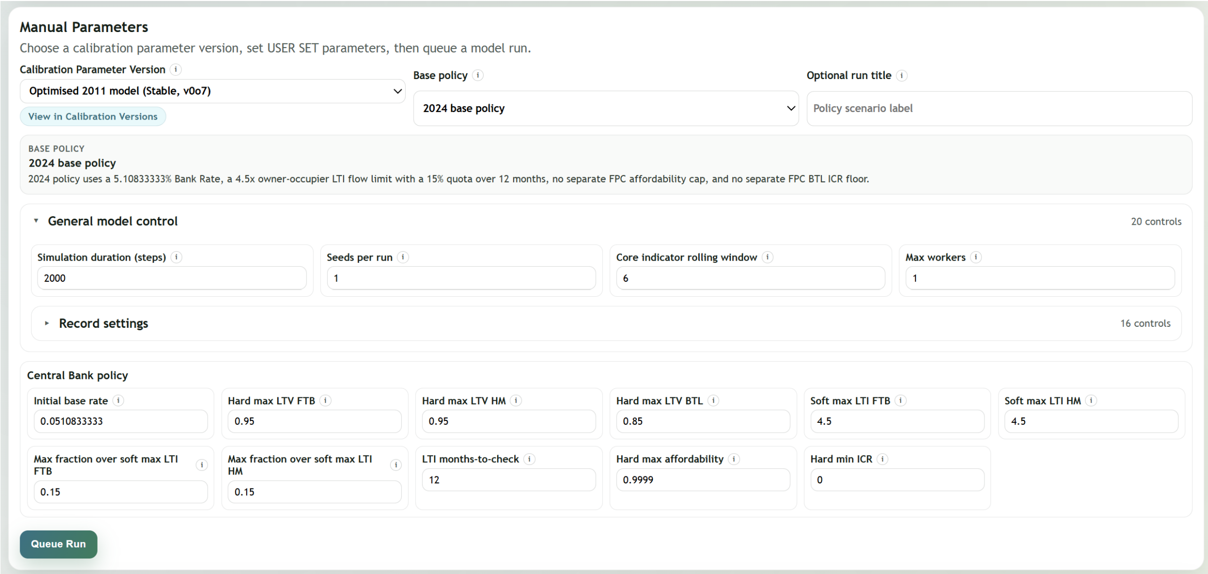

Figure D.3: Manual experiment queue run page with experiment setup in the experiment section of the publicly hosted ABM website [38].

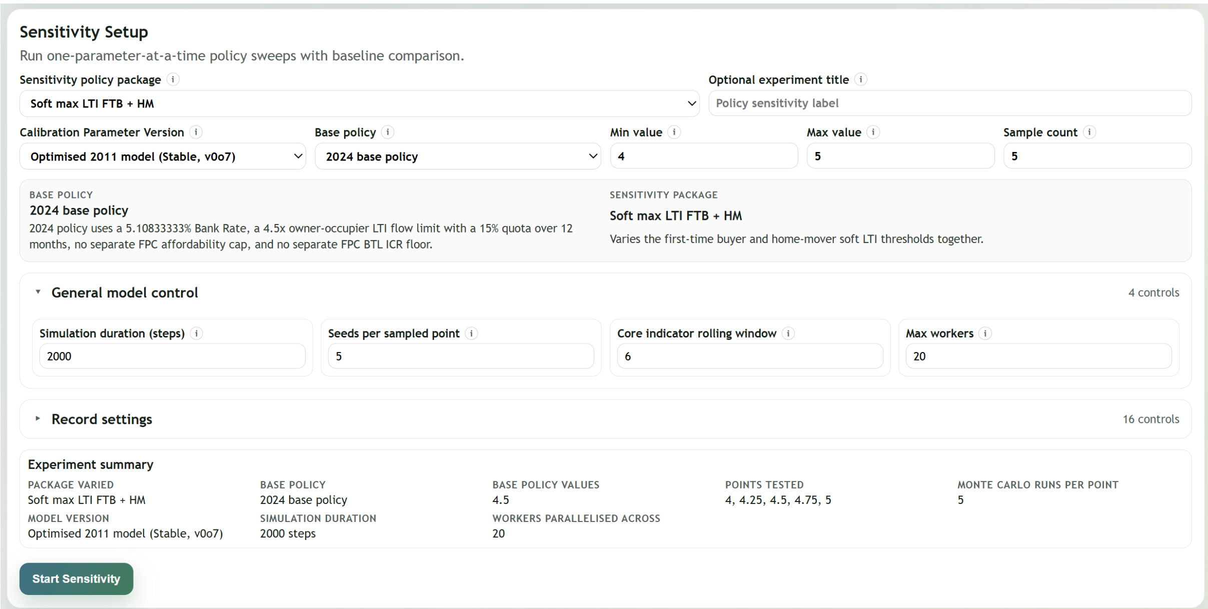

Figure D.4: Sensitivity experiment queue run page with experiment setup in the experiment section of the publicly hosted ABM website [38].

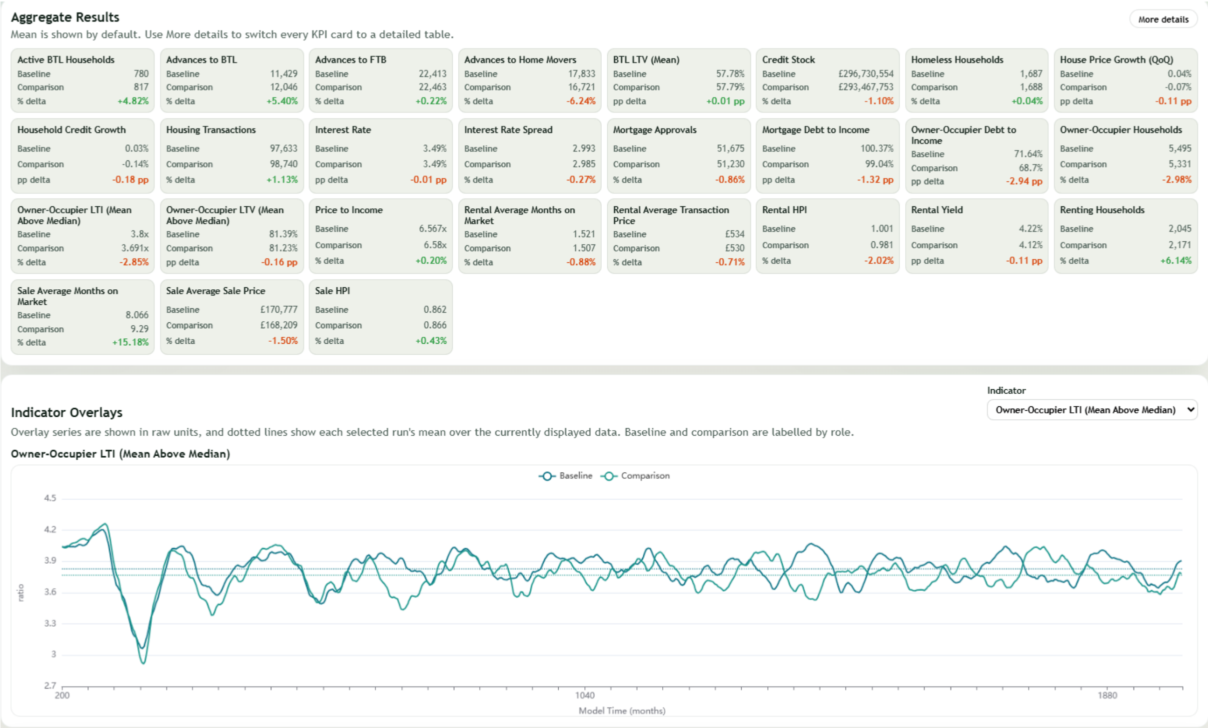

Figure D.5: Example manual experiment results in the experiment section of the publicly hosted ABM website [38].

Figure D.6: Example sensitivity experiment results in the experiment section of the publicly hosted ABM website [38].